

Auditing background-dependent inference leakage with field-level risk profiles

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Abstract

Inference leakage depends on the background information available to an adversary. A field with little predictive value alone may substantially improve inference of a sensitive target when combined with other released fields. We propose an audit that turns set-level inferability into a field-level risk profile under an explicit background budget. Budgeted marginal risk is the largest gain over admissible backgrounds relative to already-public data; background amplification isolates the part missed by single-field assessment. We establish the measure’s safety-margin interpretation and its connection to minimal unsafe combinations. To evaluate the many required field sets, an amortized estimator combines reconstruction-trained and random features with a separate ridge readout for each set. The scan proposes backgrounds, and dedicated retraining verifies selected gains and criticality evidence. Evaluation across four power-system data sets, ten targets and 124,425 exhaustively retrained field sets shows that average contextual gain is typically much smaller than its maximum. The estimator achieves a mean absolute marginal-risk error of 0.015 and a field-ranking correlation of 0.968. Verifying three backgrounds per field recovers 97.8% of total marginal risk; critical-field recall is 76–91% without false positives in the evaluated settings. At 48–64 ms per set, the estimator recovers comparable certified marginal risk to TabPFN v2 at 13–110 times lower scan cost, while their threshold-specific recall differs. A controlled dispatch case and a withholding application demonstrate how the profile explains and supports decisions about data release.

Keywords: Inference leakage, Background knowledge, Security auditing, Marginal contribution, Amortized estimation, Power-system data

1. Introduction

Data-release security depends on what a recipient can infer after combining a release with information already available. This creates a difficulty for field-level assessment: a field that predicts a sensitive target poorly in isolation may become informative in the presence of another field. An audit based only on individual correlations or prediction accuracies can therefore miss a useful inference channel.

The inference and aggregation problem has long been recognized in database security [1, 2]. Recent work on reconstruction from multi-attribute releases further illustrates the importance of relations between attributes [3]. These observations motivate a specific auditing question: how much can releasing a particular field improve supervised inference of a protected target over any admissible background? The question concerns the information gained from a combination, rather than only the predictive importance assigned to a field by one fitted model.

Power-system data provide a demanding application. Prices, forecasts, schedules and operating quantities arise from shared physical and economic constraints. Different fields are also released at different times, creating explicit public, candidate and protected roles [4]. For example, a unit’s output can identify an active offer segment; the local price may then reveal a private markup that neither field predicts well alone. Published nodal prices have also been used to recover non-public network information [5]. These relations make the domain useful for examining inference leakage with real operational structure.

Existing approaches supply parts of an answer. Inference graphs propagate sensitivity through relations between fields [6]. Feature-attribution methods convert the value of field sets into in-

dividual scores, but averages over backgrounds can dilute a rare large gain, and scores evaluated in one context can suppress substitutable fields [7–9]. The marginal contribution index addresses the maximum-gain question [9]. For release auditing, we extend its setting with an explicit public baseline and background budget, and retain the amplification and witness combinations as part of the output.

The resulting audit is computationally demanding. With 35 fields and two-field backgrounds, estimating every field’s risk requires the inferability of 7,175 sets per target. Refitting an attacker family for each set takes hours. We therefore share a nonlinear representation across subsets while fitting their readouts separately. The representation combines reconstruction-trained features with an untrained feature branch, and three independent predictions are averaged. A scan over estimated values proposes a small number of backgrounds for dedicated retraining. This links efficient search to verified evidence while also producing a quantitative risk profile for every field (Fig. 1).

Our contributions are as follows.

- *A field-level leakage profile.* We define budgeted marginal risk from set-level inferability, separate background amplification from single-field risk, and establish the safety-margin and minimal-unsafe-set connections needed to interpret the profile.
- *An efficient and verifiable audit.* We combine shared features with subset-specific adaptation and a retraining stage. Evaluation distinguishes profile error, recovered marginal risk, critical-field recall and scan cost, including comparisons with linearized, warm-started, surrogate and foundation-model alternatives.
- *An extensive application study.* Across four power-system data sets, 264 field–target pairs and 124,425 exhaustively retrained sets, we measure how often background information changes field risk. A controlled mechanism, budget sensitivity analysis and withholding application connect the numerical results to interpretable release decisions.

The next sections specify the threat model and measurement, describe the audit, and evaluate its security and computational behavior. Related work is then revisited in light of the resulting field-level evidence.

2. Threat Model and Field-Level Risk

2.1. Parties, field roles and threat model

The **auditor** is the data owner and evaluates confidentiality before release. It holds the complete history of all operational fields, including those it regards as sensitive, and grades fields offline before release. The **adversary** obtains published fields and is assumed to hold historical auxiliary labels of the sensitive target, from which it trains a supervised inference model; risk is therefore measured as out-of-sample *supervised* inferability. The adversary may combine the field under review with other fields it already holds.

Field roles are fixed by disclosure rules, not by data analysis. Let G be the candidate fields that the rules allow to be published, $B \subseteq G$ a *public baseline* that is already released ($B = \emptyset$ unless stated otherwise), $H = G \setminus B$ the fields under review, and Y_y a sensitive target. Section 4.1 instantiates these roles for four data sets from the categories and release timings of the Chinese electricity-market disclosure rules [4]. An ex-ante estimate of the *same physical quantity* as a target (e.g. a load forecast when the target is metered load) is removed from G , because it is a copy of the target rather than an inference channel.

2.2. Set-level inferability

Definition 1 (Population inferability).

$$V_y^*(S) = 1 - \inf_f \mathbb{E}[(Y_y - f(X_S))^2] / \text{Var}(Y_y),$$

the infimum taken over measurable f . It satisfies $V_y^*(\emptyset) = 0$, $0 \leq V_y^* \leq 1$ and is monotone in S .

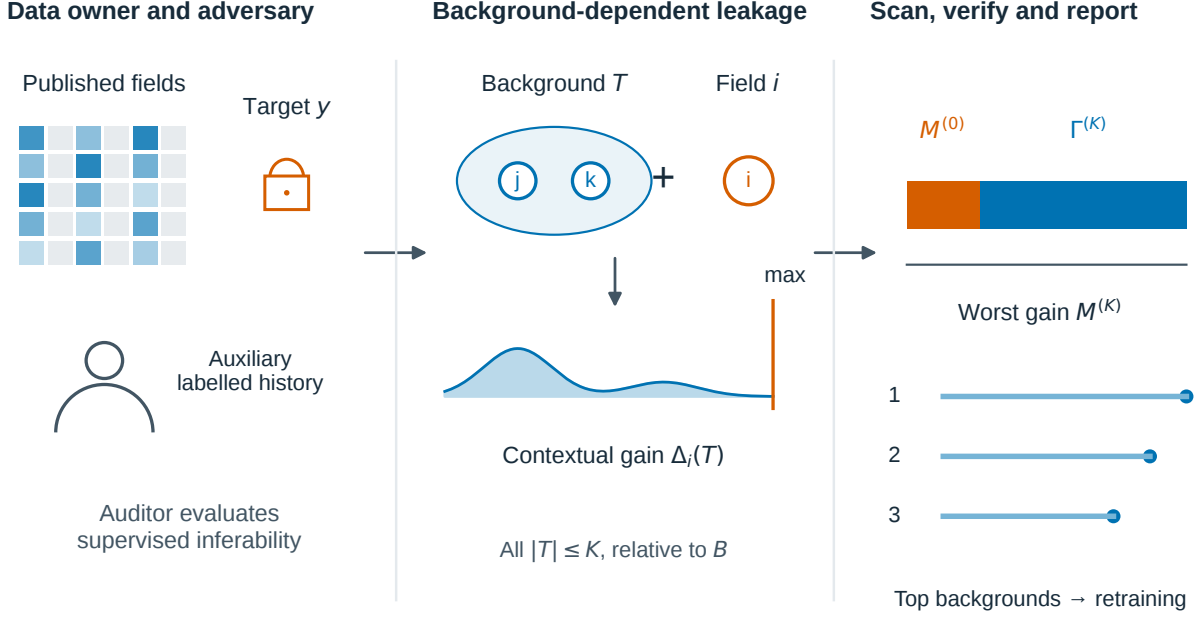


Figure 1: Audit setting and evidence flow. The data owner evaluates a candidate field against an adversary with published fields and labelled auxiliary history. Alternative admissible backgrounds produce different inferability gains. The scanner estimates the field profile and proposes backgrounds whose gains are verified by retraining. The miniature profile is illustrative.

Definition 2 (Empirical inferability). *The data are split into training and test rows, with a validation part inside training. For an attacker family $\mathcal{H}$, every $h \in \mathcal{H}$ is trained on X_S only; the attacker and its training epoch are selected on validation, and the test R^2 is reported: $V_y^{\text{ret}}(S) = R_{\text{test}}^2(\hat{h}_S)$ with $\hat{h}_S = \arg \max_{h \in \mathcal{H}, \text{epoch}} R_{\text{val}}^2(h; S)$.*

An adversary holding a set can also use any of its subsets. We therefore evaluate the subset-closed inferability

$$V_y(S) = \max_{T \subseteq S} [V_y^{\text{ret}}(T)]_+, \quad V_y(\emptyset) = 0. \quad (1)$$

This retraining-based reference defines the empirical risk tables used throughout the paper.

2.3. Contextual gain

The quantity from which everything else follows is the gain that releasing field i brings over a given background T of fields the adversary already holds,

$$\Delta_i(A) = V_y(A \cup \{i\}) - V_y(A), \quad A = B \cup T. \quad (2)$$

$\Delta_i(B)$ is the single-field risk. The same field can be nearly useless over one background and decisive over another: a price series says little about a unit’s output alone, but together with the unit’s forecast availability it reveals the residual quantity the unit covers. A field-level risk quantity must therefore summarize a whole *distribution* of contextual gains $\{\Delta_i(B \cup T)\}_T$, and the question is which summary serves security.

2.4. Budgeted marginal risk and background amplification

Definition 3 (Budgeted marginal risk, background amplification). *For target y , public baseline B , fields under review $H = G \setminus B$ and budget K ,*

$$M_{i \rightarrow y}^{(K)} = \max_{T \subseteq H \setminus \{i\}, |T| \leq K} \Delta_i(B \cup T), \quad (3)$$

$$\Gamma_{i \rightarrow y}^{(K)} = M_{i \rightarrow y}^{(K)} - M_{i \rightarrow y}^{(0)} \geq 0. \quad (4)$$

Any maximizer T^* is a witness background.

$M_{i \rightarrow y}^{(K)}$ is the largest increase of inferability that releasing i can cause when the adversary holds at most K other fields; $\Gamma_{i \rightarrow y}^{(K)}$ is the part of it that exists only because of the background, i.e. the amplification of single-field risk by combination. For each (field, target) pair the audit reports the profile $(M_{i \rightarrow y}^{(0)}, \Gamma_{i \rightarrow y}^{(K)}, M_{i \rightarrow y}^{(K)})$ together with the top- r backgrounds ranked by Δ_i ; in our data the top backgrounds are often nearly tied and share a core partner field, so the partner is more informative than a single argmax. At $K = 1$ the amplification has a closed form.

Proposition 1 (Amplification at $K = 1$). *With $I_{ij}^B = V_y(B \cup \{i, j\}) - V_y(B \cup \{i\}) - V_y(B \cup \{j\}) + V_y(B)$, $\Gamma_{i \rightarrow y}^{(1)} = \max(0, \max_j I_{ij}^B)$.*

Proof. Over the background $\{j\}$, $\Delta_i(B \cup \{j\}) = M_{i \rightarrow y}^{(0)} + I_{ij}^B$; over $\emptyset$ it is $M_{i \rightarrow y}^{(0)}$. Take the maximum. $\square$

A field is amplified by one background field exactly when it is strictly super-additive with it, and the amplification equals the strongest positive interaction.

2.5. Relation to feature importance

An average of contextual gains, as in Shapley values or SAGE [7], describes typical contribution. The maximum asks whether an admissible background produces a large gain, which is the relevant question when reviewing a release. In our data the median ratio of the average gain to this maximum is 0.32 (Section 5.1). The MCI [9] formalizes this maximum contribution; the public baseline and background budget specialize it to the audit setting. Appendix G compares the requirements met by alternative field scores.

Proposition 2 (Relation to MCI). *Let $v_B(T) := V_y(B \cup T) - V_y(B)$ for $T \subseteq H$. Then v_B is monotone, non-negative and $v_B(\emptyset) = 0$; $M_{i \rightarrow y}^{(K)} = \max_{|T| \leq K} [v_B(T \cup \{i\}) - v_B(T)]$ is the budget-constrained MCI of v_B ; for $K \geq |H| - 1$ and $B = \emptyset$ it is the MCI of V_y .*

The axiomatic properties of MCI for monotone evaluation functions [9]—dummy, symmetry, the self-contribution lower bound $M_{i \rightarrow y}^{(K)} \geq M_{i \rightarrow y}^{(0)}$ and invariance to duplicated features—therefore transfer and are cited rather than claimed; $M_{i \rightarrow y}^{(K)}$ is non-decreasing in K and reaches the MCI at $K = |H| - 1$.

2.6. Security meaning

Theorem 1 (Threshold-free safety margin). *Fix i, y, B, K . A number $m \geq 0$ satisfies “for every threshold τ and every background $|T| \leq K$: $V_y(B \cup T) \leq \tau - m \Rightarrow V_y(B \cup T \cup \{i\}) \leq \tau$ ” if and only if $m \geq M_{i \rightarrow y}^{(K)}$. For a fixed τ , the smallest m with this property can be strictly smaller than $M_{i \rightarrow y}^{(K)}$.*

Proof. If $m \geq M_{i \rightarrow y}^{(K)}$ then $V_y(B \cup T \cup \{i\}) = V_y(B \cup T) + \Delta_i(B \cup T) \leq (\tau - m) + m$. Conversely, for any T choose $\tau = V_y(B \cup T) + m$; the property gives $\Delta_i(B \cup T) \leq m$. Strictness: with $B = \emptyset$, $K = 1$, $V_y(\{i\}) = 0.1$, $V_y(\{j\}) = 0.5$, $V_y(\{i, j\}) = 0.9$ one has $M_{i \rightarrow y}^{(1)} = 0.4$, yet at $\tau = 0.95$ the necessary margin is 0. $\square$

Thus $M_{i \rightarrow y}^{(K)}$ is the answer to “by how much can publishing i raise the adversary’s accuracy, whatever it already holds within budget”—a property of the field that does not depend on the threshold later chosen for grading.

Definition 4 (Criticality and minimal unsafe sets). *Under baseline B , $E \subseteq H$ is a τ -minimal unsafe set (MUS) if $V_y(B \cup E) > \tau$ and $V_y(B \cup E') \leq \tau$ for every $E' \subsetneq E$. Field i is K -background τ -critical if some T with $|T| \leq K$, $i \notin T$, has $V_y(B \cup T) \leq \tau < V_y(B \cup T \cup \{i\})$.*

Theorem 2 (Criticality $\iff$ small MUS). *If V_y is monotone and $V_y(B) \leq \tau$, field i is K -background τ -critical if and only if it belongs to a MUS of size at most $K + 1$.*

Proof. ($\Leftarrow$) For a MUS $E \ni i$ with $|E| \leq K + 1$ take $T = E \setminus \{i\}$. ($\Rightarrow$) If $V_y(B \cup T) \leq \tau < V_y(B \cup T \cup \{i\})$, take an inclusion-minimal unsafe $E \subseteq T \cup \{i\}$; it is a MUS of size $\leq K + 1$. If $i \notin E$ then $E \subseteq T$ and monotonicity gives $V_y(B \cup T) \geq V_y(B \cup E) > \tau$, a contradiction. $\square$

Theorem 2 makes the field-level flag a faithful projection of the set-level danger structure: flagging the τ -critical fields misses no field that takes part in a dangerous combination of at most $K + 1$ fields, and a threshold-crossing background identifies the combination. The same list of minimal unsafe sets also supports withholding decisions (Section 6).

2.7. Background budget

The budget K specifies how many additional fields the adversary may combine with the field under review, beyond the public baseline. We use $K = 2$ for the main audit and evaluate $K = 3$ on every data set. Small budgets capture interpretable combinations and control the enumeration cost. Section 5.5 reports both changes in marginal risk and changes in threshold crossings as the budget increases.

3. Amortized Scanning with Retraining-Based Verification

The auditor must evaluate many subsets under the same target, historical data and public baseline. This produces a collection of closely related prediction tasks. The estimator shares their nonlinear representation and adapts the readout to each set. Its output serves two purposes: estimating the field profile and proposing backgrounds for verification (Fig. 2).

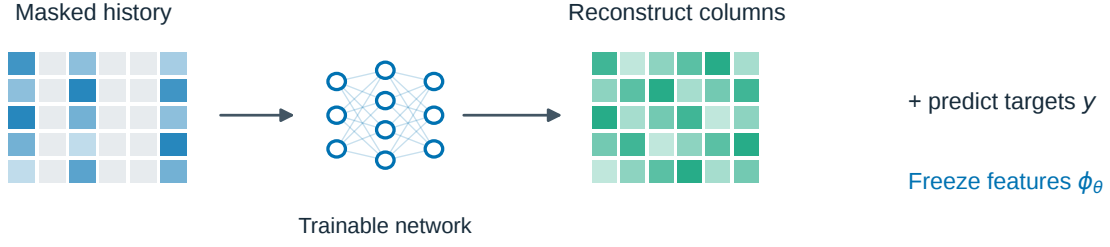
3.1. Masked pre-training

The adversary does not know in advance which fields it will obtain. We model this by pre-training one network under random visibility masks. A mask $m \in \{0, 1\}^p$ is drawn by first sampling the number of visible fields uniformly and then a uniform subset of that size; the network $f_\theta(x \odot m, m)$ receives the masked values and the mask itself, so that a hidden field is distinguishable from a zero value. The objective combines predicting the sensitive targets from the visible fields with reconstructing every hidden column from the visible ones [10, 11],

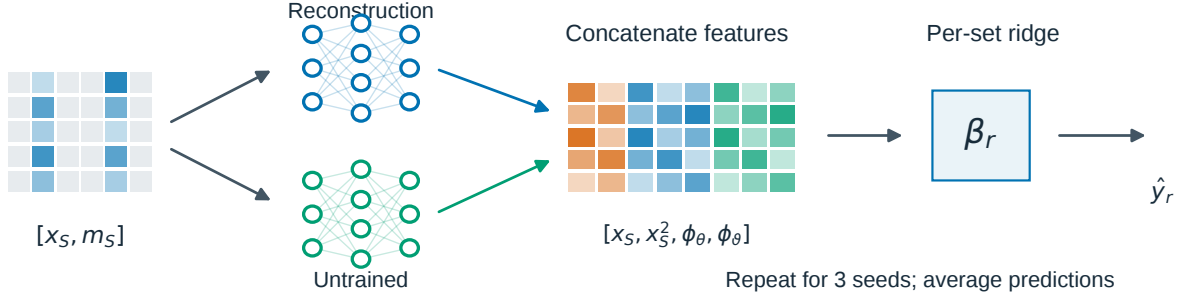
$$\mathcal{L}(\theta) = \mathbb{E}_m \left[\|f_\theta^y(x \odot m, m) - y\|^2 + \gamma \frac{\|(1 - m) \odot (f_\theta^x(x \odot m, m) - x)\|^2}{\|1 - m\|_1} \right], \quad (5)$$

with $\gamma = 1$. Reconstruction broadens the representation beyond the few directions needed to predict the selected targets. It encourages features that describe the operating state shared by the observed columns. Section 5.3 evaluates this contribution; Appendix C examines reuse across target definitions. The last hidden layer (256 units) is kept as a frozen feature map $\phi_\theta(x \odot m, m)$; early stopping uses the target loss on validation rows only. Alongside each pre-trained backbone we keep an untrained network of the same architecture, ϕ_ϑ with random initial weights ϑ . The two play different roles in the readout: the pre-trained features supply target-aligned directions and the operating state, the random features the high-dimensional bulk of a kernel regression [12] (Section 5.3).

(a) Learn a shared representation once



(b) Adapt to each field set S



(c) Scan backgrounds → Retrain top candidates → Verify gains

Figure 2: The amortized attacker. (a) One backbone is pre-trained under random visibility masks on the auditor’s data, to predict the targets and reconstruct the hidden columns, and frozen. (b) A query (S, y) solves a ridge readout in closed form on the features of S —from the pre-trained backbone and an untrained network of the same architecture—per seed, and averages the predictions. (c) The scan proposes backgrounds per field; retraining only these certifies a lower bound on $M_{i \rightarrow y}^{(K)}$.

3.2. Subset-specific closed-form readout

For a query (S, y) with mask m_S and seed r , the feature block is

$$Z_r(S) = [x \odot m_S, (x \odot m_S)^2, \phi_{\theta_r}(x \odot m_S, m_S), \phi_{\vartheta_r}(x \odot m_S, m_S)], \quad (6)$$

standardized with training statistics. A ridge readout

$$\beta_r(S) = (Z_r^\top Z_r + \lambda I)^{-1} Z_r^\top y \quad (7)$$

is solved in closed form on the training rows, with λ chosen by five-fold cross-validation inside the training rows. Each of $R = 3$ seeds (a pre-trained and an untrained network) has its own readout—three separate closed-form solves per query, not one solve on the features of all seeds—and their predictions are averaged, $\hat{y} = R^{-1} \sum_r Z_r^{\text{test}} \beta_r(S)$, and $\hat{V}_y(S) = R_{\text{test}}^2(\hat{y})$, followed by the monotone closure. Three safeguards stabilize the readout: a floor of 10^{-3} on feature standard deviations, clipping test features to their training range, and clipping predictions to the target’s training range. All bounds are obtained from training rows. These operations limit large extrapolation errors that could otherwise dominate a maximum of estimated marginal gains.

We use the three-seed reconstruction-plus-random configuration throughout the main evaluation. Its 48–64 ms cost per field set is suitable for the audit workloads studied here, while its lower profile error supports both grading and background selection. The reconstruction-only configuration offers a lower-cost option for larger workloads (Section 5.3).

3.3. Why the readout is solved per subset

A cheaper alternative lets the network output the prediction directly, $\hat{y} = f_\theta(x \odot m_S, m_S)$, and reads $\hat{V}$ off it. Then one set of output weights serves every conditioning pattern, i.e. the attacker is forced to combine every subset of fields in the same way—the very effect that combination risk is about. Solving the readout per subset keeps the expensive representation shared and the combination rule specific.

3.4. Error propagation

Write the estimate as $\hat{V}(S) = V(S) + b + \varepsilon(S)$, where b is a bias common to the sets of interest and $\varepsilon(S)$ the set-specific part. Since a marginal is a difference of two nested sets, $\hat{\Delta}_i(T) - \Delta_i(T) = \varepsilon(T \cup \{i\}) - \varepsilon(T)$ and

$$|\hat{M}_{i \rightarrow y}^{(K)} - M_{i \rightarrow y}^{(K)}| \leq \max_{|T| \leq K} |\varepsilon(T \cup \{i\}) - \varepsilon(T)|, \quad (8)$$

independently of b . A readout on a frozen representation typically underestimates the fully trained attacker by a similar amount on nested sets, which is exactly a common-mode b ; this is why an estimator can be biased in V yet accurate in M . The quantity to control is the *marginal* error, which we report throughout.

3.5. Scan and certify

The estimator is used to *scan*: it produces $\hat{M}^{(K)}$ for every field together with a ranked list of candidate backgrounds. Decisions are then *certified*: for each field, dedicated attackers are retrained on the top- k backgrounds only, and $L_{i \rightarrow y}^{(K)} = \max_{T \in \text{top-}k} \Delta_i(B \cup T) \leq M_{i \rightarrow y}^{(K)}$ is a lower bound computed entirely from retrained attackers. The ratio $\sum_i L / \sum_i M$ measures how much of the true risk the scan recovers at a certification cost of k retrained backgrounds per field instead of all $\binom{|H|-1}{\leq K}$.

4. Experimental Setup

4.1. Data sets and rule-based field roles

Table 1 summarizes the four data sets. In all of them, candidate fields are information that the disclosure rules [4] release *before* the target, and targets are either specific information (unit output, Art. 17(3)), quantities not on the disclosure list at hourly resolution (hourly line flow; only daily averages of important lines are listed), or actual operating quantities released on the following day (actual load, total generation, tie-line exchange; Appendix items 6.57–6.66). Using the release timing to assign roles does not involve any temporal split of the data; all splits are random.

RTS-GMLC. The IEEE Reliability Test System 2019 update [13] provides a 73-bus network with one year of realistic hourly load, wind, solar and hydro series. We clear an hourly DC optimal power flow with piecewise-linear offers built from the published heat-rate curves (8,784 hours, 26.7% congested). Candidates are the 23 fields a market would publish early (zonal load forecasts, renewable and hydro forecasts, reserve requirements, system energy price and nine nodal prices, three congestion indicators); targets are the outputs of a coal and a combined-cycle unit and the hourly flow on internal line C35.

NEM. One year (2024) of the Australian National Electricity Market [14], averaged to hours. Candidates are regional demand, available capacity, semi-scheduled wind and solar forecasts, regional prices and interconnector flows (30 fields); targets are the actual outputs of a coal unit (BW01), a hydro unit (TUMUT3) and a gas combined-cycle unit (PPCCGT). Unit output is public in the NEM but is specific information under the Chinese rules; this data set supplies observed unit outputs for the stated disclosure scenario.

PJM and CAISO. System-level hourly data for one year. Targets are the actual quantities released on the next day: PJM metered load, total generation and net actual interchange, and CAISO actual load. Candidates are forecasts, day-ahead and real-time prices and their components, ancillary-service requirements and cleared volumes, and schedules (22–35 fields). Fields released on

Table 1: Audit data and reference coverage. Rows give training/test sample sizes. Each reference table contains all nonempty field sets of size at most four. The field-role basis is described in the text and Table B.8.

Data	Targets	Train/test	Fields	Sets
RTS-GMLC	Units 223, 221; line C35	6149/2635	23	10,902
NEM 2024	BW01, TUMUT3, PPCCGT	6149/2635	30	31,930
PJM load	Metered load	4589/1967	22	9,108
PJM gen./interch.	Generation; net interchange	4589/1967	24	12,950
CAISO	Actual load	6105/2617	35	59,535

the same day as the target (fuel-mix generation, actual renewable output, other balancing areas’ actual load) are excluded, as are the load forecasts of the target area, which are ex-ante copies of the target (Section 2). Near-copies that are distinct quantities (e.g. sub-area forecasts for CAISO load) are kept; removing them is analysed as a sensitivity case.

Controlled mechanism. A fifth data set is built on the RTS-GMLC market with a private hourly offer markup of one unit as the target and eleven fields chosen to span the known mechanism (Section 5.4); all 2,047 of its field sets are retrained.

4.2. Ground truth

All variables are standardized with training statistics; 70/30 random train/test splits are used, with 15% of the training rows as validation. The attacker family has three members: a single-target multilayer perceptron (two hidden layers of 128 units, dropout 0.15, Adam, early stopping on validation), a multi-target perceptron trained on all targets of a data set, and histogram gradient boosting with two configurations. For every set of up to four fields, every member is trained on that set alone; the member and epoch are chosen on validation and the test R^2 is reported, followed by the monotone closure (1). This amounts to 124,425 field sets and several hundred thousand trained models. These exhaustive tables provide the reference for evaluating the estimators and field scores.

4.3. Baselines

Field scores compared with $M^{(2)}$ are: Pearson r^2 , squared Spearman correlation, mutual information [15], distance correlation [16], single-field retraining $M^{(0)}$ (current practice), LOCO [8], permutation importance [17], SAGE with marginal imputation [7], and an inference-graph score that follows the propagation idea of [6] without its expert inputs: edge weights are the single-field predictability of one field from another (gradient boosting, validation R^2), the target node carries risk one, and risk is propagated over three hops with decay 0.5. All baseline scores are computed on training rows only.

Estimators compared with ours avoid retraining the attacker family for each set in different ways: per-set linear and quadratic regression (retraining the simplest attackers); a model trained on all fields with the removed fields replaced by their mean [18], linearized around its parameters with a ridge correction per set (LazyVI [19]), or warm-started and trained with early stopping per set [20]; a surrogate trained under random masks whose output is read directly [21, 22]; and the tabular foundation model TabPFN v2 [23] used in context. Design choices of our estimator—the feature map, the use of seeds, the safeguards and the pre-training masks—are compared separately (Table E.10); details of all implementations are given in Appendix B.

4.4. Metrics

For set values: mean absolute error of $\hat{V}$. For field risk: mean absolute error and bias of $\hat{M}^{(2)}$, Spearman correlation across fields, and the certified ratio $\sum_i L_{i \rightarrow y}^{(2)} / \sum_i M_{i \rightarrow y}^{(2)}$ when k backgrounds per field are retrained. For decisions at thresholds $\tau \in \{0.5, 0.7\}$: recall and precision of critical fields; the number of fields withheld and the number of minimal unsafe sets of size ≤ 3 left unbroken.

Table 2: Fields exposed by single-field and combination screening. V_1 : best single field; $V_{\leq 3}$: best set of at most three fields; alone: fields with $V(\{i\}) > \tau$; comb.: fields safe alone but τ -critical with at most two background fields. Minimal unsafe sets and withholding counts are reported in Table H.13.

Target	p	V_1	$V_{\leq 3}$	$\tau = 0.5$		$\tau = 0.7$	
				alone	comb.	alone	comb.
RTS unit 223	23	1.00	1.00	10	13	10	6
RTS unit 221	23	0.99	0.99	10	13	10	9
RTS line C35	23	0.52	0.86	4	19	0	21
NEM BW01	30	0.49	0.65	0	30	0	0
NEM TUMUT3	30	0.38	0.53	0	22	0	0
NEM PPCCGT	30	0.55	0.80	2	28	0	10
PJM load	22	0.48	0.80	0	22	0	14
PJM total gen.	24	0.97	0.99	2	22	2	18
PJM net interch.	24	0.41	0.71	0	24	0	3
CAISO load	35	0.82	0.93	5	30	3	32

5. Evaluation Results

5.1. Background Amplification in Real Data

Contextual gains are widely spread. Fig. 3(b) plots, for the most amplified field of six targets, the gain $\Delta_i(T)$ over every background of at most two fields. The spread is large, and the maximum typically lies far from the bulk: the South Australian wind forecast has gain 0.19 alone, 0.20 on average over backgrounds, and 0.60 over its worst background (regional available capacity and solar forecast, which together with wind give the residual demand the gas unit PPCCGT covers). Over all 264 field–target pairs, the median ratio of the average contextual gain to the maximum is 0.32: an average-based score would report about a third of the worst-case risk.

Amplification is common. Averaged over the 264 pairs, $\Gamma^{(1)} = 0.082$, $\Gamma^{(2)} = 0.110$ and $\Gamma^{(3)} = 0.119$; one background field supplies 75% of the total amplification at $K = 2$, and $K = 2$ supplies 93% of that at $K = 3$. In 91 pairs the single-field risk is below 0.3 while the amplification exceeds 0.1 (Fig. 3a)—fields that look unremarkable alone and become informative in combination. The largest amplifications are the SA wind forecast for PPCCGT (0.19 $\rightarrow$ 0.60), the SP15 day-ahead energy price for CAISO load (0.14 $\rightarrow$ 0.51, with the NP15 wind and SP15 solar forecasts: price and renewables determine net load), and the SA available capacity for PPCCGT (0.02 $\rightarrow$ 0.36). On every pair, $\Gamma^{(1)}$ equals the strongest positive pairwise interaction (Proposition 1) to numerical precision.

Dangerous backgrounds come in families. The top backgrounds of a field are often nearly tied and share a core partner. For the SA wind forecast the five best backgrounds reach gains between 0.545 and 0.603, and all contain the SA available capacity. We therefore report the top- r backgrounds rather than a single witness; the shared partner is the actionable information (“do not co-release with”).

The witnesses are physically meaningful. Fig. 5 shows the field profile of RTS line C35 and the gain matrix of NEM unit PPCCGT. The largest increases come from combinations that reconstruct a residual quantity; for instance, the price at RTS bus 303 rises from 0.36 to 0.65 for the C35 flow together with a zonal load forecast and the congestion indicator of the neighbouring line C6 (further examples follow below). The gain matrix also shows that the same field is harmless next to some partners and decisive next to others, which a grade that ignores the background cannot represent.

Single-field grading misses most of the risk. We first ask how much of the risk in the four data sets is invisible to single-field grading. Table 2 and Fig. 4 report, for each of the ten targets, how many candidate fields are unsafe on their own and how many become critical only in combination.

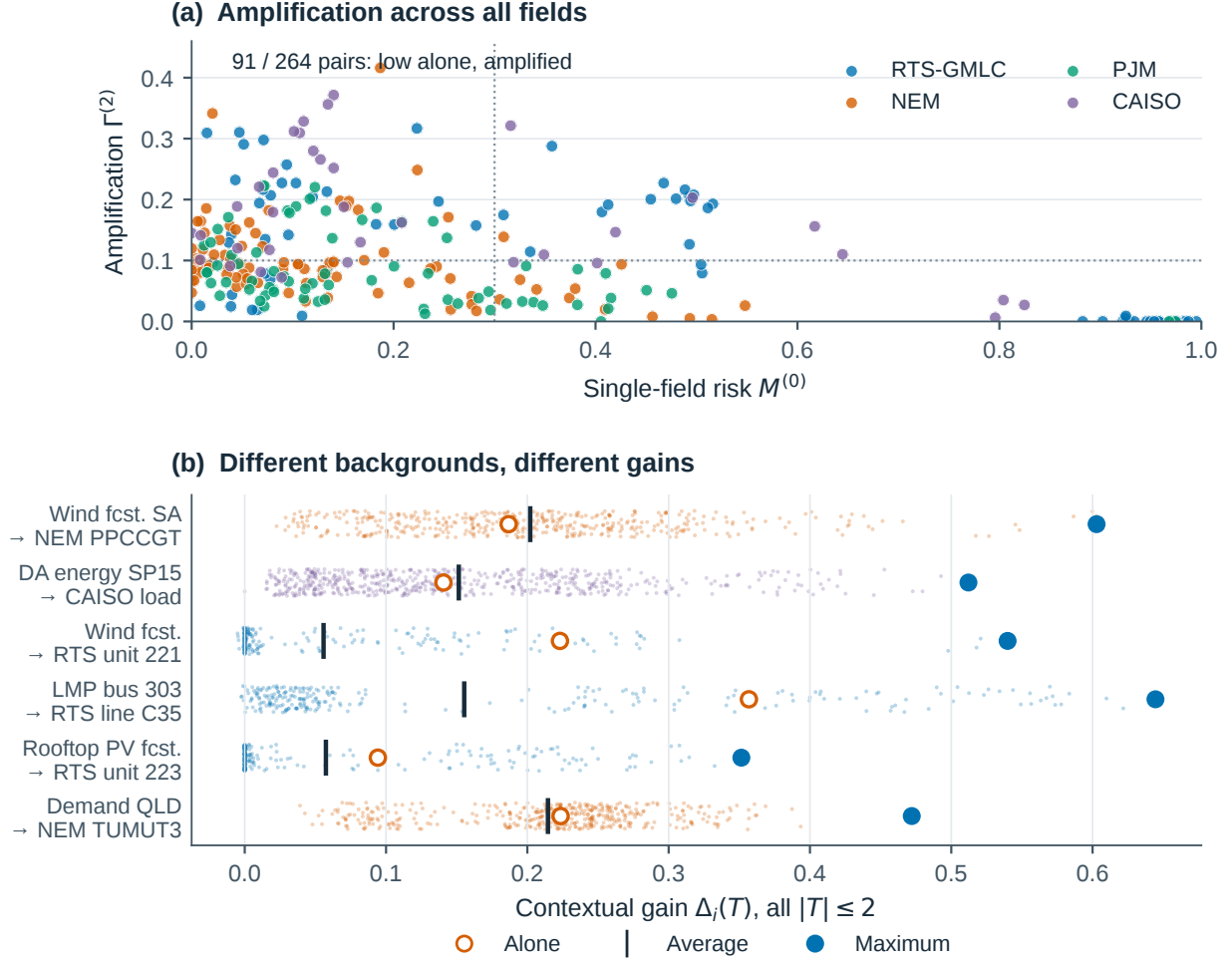


Figure 3: Background amplification over ten targets. (a) Single-field risk versus amplification $\Gamma^{(2)}$ for every field-target pair. (b) Contextual gains of the most amplified field of six targets over all backgrounds of at most two fields, with the single-field value, the average over backgrounds and the maximum.

Of the 264 field-target pairs, 256 are critical at $\tau = 0.5$ with at most two background fields, and 223 of these are safe on their own; at $\tau = 0.7$ the figures are 138 and 113. For four targets (NEM BW01 and TUMUT3, PJM load and PJM net interchange) no candidate exceeds $\tau = 0.5$ alone, yet 22 to 30 candidates take part in a dangerous combination; at $\tau = 0.7$ the same holds for RTS line C35, where 21 of 23 fields are critical and none is unsafe alone. Withholding every field that is unsafe alone breaks almost none of the minimal unsafe sets (Table H.13): 1,766 of 1,799 at $\tau = 0.5$ and 534 of 559 at $\tau = 0.7$ remain.

Where single fields do leak, combinations still dominate. Some targets are exposed by a single close proxy: RTS unit outputs are almost determined by a nodal price ($V_1 = 0.99$), and PJM total generation by the system load forecast (0.97). For these, $M^{(0)}$ already flags the proxy, and the remaining risk is again combinatorial (e.g. 22 of 24 PJM fields are combination-critical). Removing such near-proxies from the candidate pool (Section 4.1) lowers the best single field to 0.42 for both PJM total generation and CAISO load but leaves every remaining field combination-critical at $\tau = 0.5$ (22/22 and 26/26), so the conclusion does not depend on where the proxy boundary is drawn.

5.2. Comparison of Field-Level Risk Scores

The mechanism case (Section 5.4) already shows how each family of scores fails on a known structure; a purely synthetic example with the same three failure modes—synergy, redundancy and

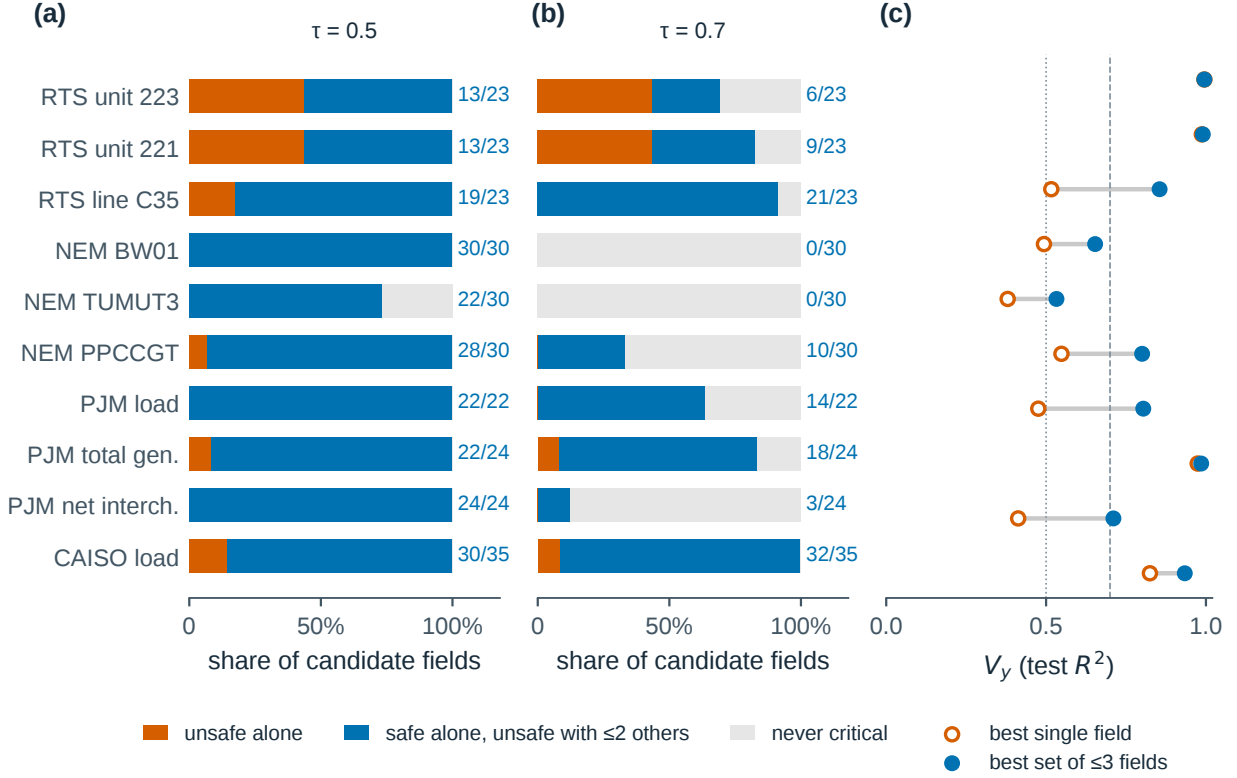


Figure 4: Share of candidate fields that are unsafe alone (orange), safe alone but critical with at most two background fields (blue), or never critical (grey), at (a) $\tau = 0.5$ and (b) $\tau = 0.7$; numbers give combination-critical fields over all candidates. (c) Best single field versus best set of at most three fields.

indirect contribution—is given in Appendix D. Here we compare the scores as rankings of the fields of the ten real targets, against critical fields defined by retrained ground truth.

Ranking. Table 3 compares the scores as rankings. At $\tau = 0.7$, $\hat{M}^{(2)}$, its certified version and the exact $M^{(2)}$ share the highest R-precision (0.894, against 0.866 for mutual information, 0.830 for the inference graph, 0.803 for single-field retraining, 0.769 for SAGE and 0.727 for LOCO), and withholding fields in the order of certified $M^{(2)}$ exceeds the optimum by only 0.3 fields (exact $M^{(2)}$: 0.4). At $\tau = 0.5$ almost every field is critical and every score reaches an R-precision of about 0.98; here single-field retraining ranks as well as $M^{(2)}$. As a *ranking*, single-field risk is thus a strong baseline when most fields are dangerous; the advantage of $M_{i \rightarrow y}^{(K)}$ lies in the decisions it supports and at high thresholds. Attribution scores computed on one fitted model (LOCO, permutation importance, SAGE) are the weakest throughout, because they measure what that model uses rather than what an adversary can infer.

5.3. The Amortized Attacker

Fidelity. Fig. 6 compares the amortized estimates with retrained ground truth for all sets of size ≤ 3 . The estimate of V is biased downward, visibly so for PJM, where the training set is smallest (bias -0.03 to -0.04); averaged over the ten targets the error is 0.018 with a signed bias of -0.013 , i.e. largely common-mode. As predicted by (8), the field-level risk is estimated more accurately than the set values it is built from: the mean absolute error of $\hat{M}^{(2)}$ is 0.015 and its Spearman correlation with the exact $M^{(2)}$ across fields is 0.968 (Table 5). Certification recovers most of the exact risk from very few retrained backgrounds: retraining the single best-scanned background per field yields a lower bound equal to 96.1% of the exact $M^{(2)}$ on average, three backgrounds yield 97.8%, and on RTS-GMLC and CAISO the top-3 ratio exceeds 99% (Fig. 6c). For a field set of 35

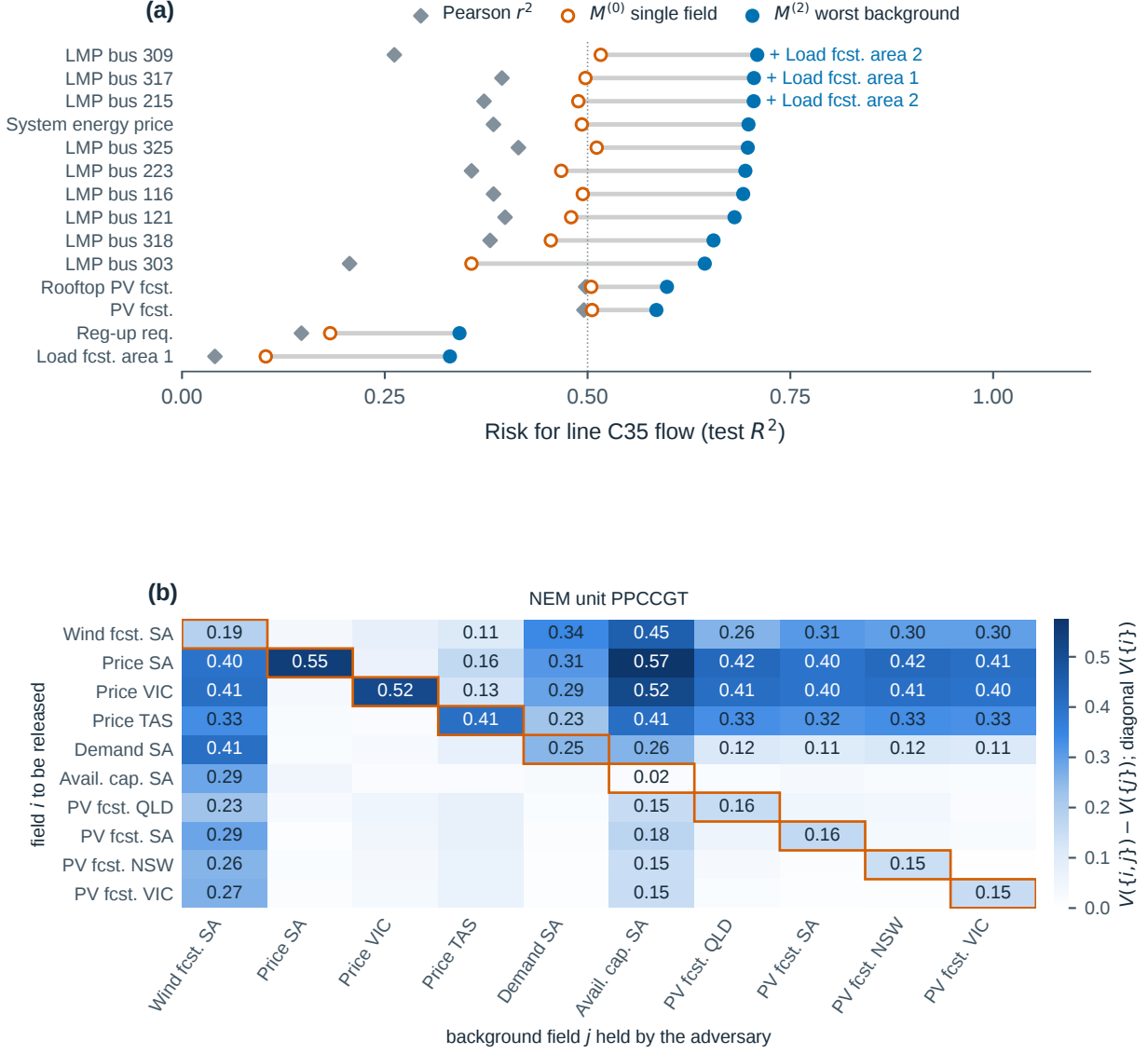


Figure 5: (a) Field profile for the hourly flow on RTS line C35 (14 fields with the largest $M^{(2)}$): Pearson r^2 , single-field risk and worst-background risk, with the witness background for the three riskiest fields. (b) Gain matrix for NEM unit PPCCGT: entry (i, j) is $V(\{i, j\}) - V(\{j\})$, the diagonal is $V(\{i\})$; values below 0.10 are not printed.

candidates this replaces the $1 + 34 + \binom{34}{2} = 596$ backgrounds per field of an exhaustive evaluation by three.

Amplification and dangerous backgrounds. The profile of Section 5.1 is estimated with similar accuracy: the mean absolute error of $\hat{\Gamma}^{(2)}$ is 0.014. Identifying the single worst background is harder—the best-scanned background is the worst one for 55% of the fields (23–87% across targets), and the worst one is among the three best-scanned for 74% and among the five best-scanned for 79%—but this matters little, because dangerous backgrounds come in families of nearly tied members: retraining the three best-scanned backgrounds per field recovers 94.8% of the total amplification.

Comparison with other estimators. Table 4 and Fig. 7 compare the estimator with other ways of obtaining the inferability of many field sets without retraining the attacker family for each; the implementations and settings are listed in Appendix B. *Re-using a model trained on all fields.* Replacing the removed fields by their mean [18] collapses (error of V 0.44): a network trained on all fields cannot predict from three of them. Linearizing it around its parameters and fitting a

Table 3: Ranking quality over the ten targets. R-precision: share of true critical fields among the top- c fields, c being the true number of critical fields (a threshold that favours the baselines). Excess: fields withheld in score order until no MUS remains, minus the optimum.

Score	R-precision		excess withheld	
	$\tau = 0.5$	$\tau = 0.7$	$\tau = 0.5$	$\tau = 0.7$
Pearson r^2	0.977	0.815	5.4	3.0
Spearman ρ^2	0.977	0.767	3.5	2.8
Mutual information	0.977	0.866	2.0	2.1
Distance correlation	0.982	0.833	3.0	2.0
Single-field $V(\{i\})$	0.982	0.803	1.6	1.9
LOCO	0.977	0.727	11.0	8.0
Permutation importance	0.968	0.795	8.9	4.7
SAGE	0.968	0.769	8.6	6.3
Inference-graph propagation	0.982	0.830	5.6	2.7
$\hat{M}^{(2)}$ (amortized, no retraining)	0.982	0.894	2.3	0.9
$M^{(2)}$ scan-certify ($k = 3$)	0.982	0.894	1.5	0.3
$M^{(2)}$ exact	0.977	0.894	1.5	0.4

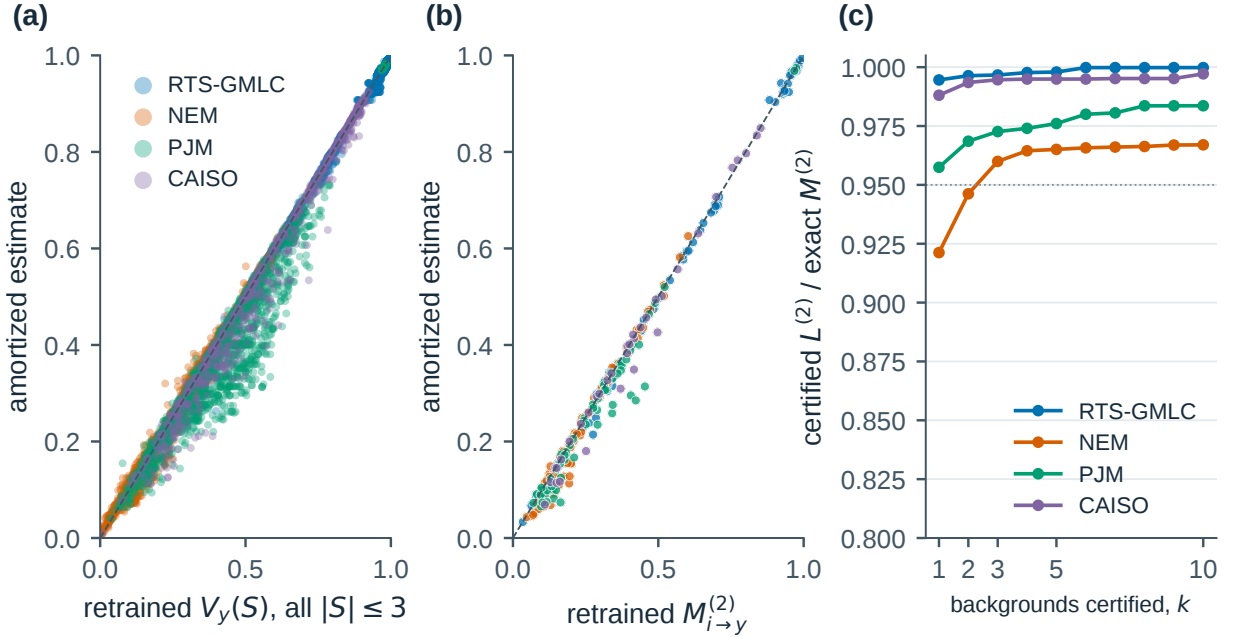


Figure 6: Amortized estimates against retrained ground truth. (a) $\hat{V}_y(S)$ for sets of size ≤ 3 (random subsample). (b) $\hat{M}^{(2)}$ for every field and target. (c) Certified lower bound over exact $M^{(2)}$ when the top- k scanned backgrounds per field are retrained.

ridge correction per set (LazyVI [19]) is accurate (0.037 on 200 sets per data set) but costs 1.5–9.1s per set in the double precision it needs, so that a complete $K = 2$ table would take one to eleven hours per data set; we therefore ran it on these 200 sets only. Warm-starting from the full model and training with early stopping [20] costs 0.18s per set and reaches an $M^{(2)}$ error of 0.047.

Table 4: Accuracy and scan cost of alternative estimators, averaged over ten targets. MAEs use all sets of at most three fields. Cert.@1 is the recovered marginal-risk ratio after verifying one background per field. Recall is scan-certify critical-field recall at $\tau = 0.7$; precision is 1.00 for all methods. Times cover all targets of one data set. LazyVI was evaluated on 200 sets only (Fig. 7). Additional metrics are in Appendix Appendix F.

Estimator	MAE V	MAE $M^{(2)}$	cert.@1	Recall	Time/set
Retraining (reference)	0	0	1	1.00	14–56 s
Linear	.212	.144	.803	.16	0.7–1.0 ms
Quadratic	.132	.096	.859	.27	0.9–1.6 ms
Mean substitution	.442	.243	.295	.00	0.1 ms
Warm start	.066	.047	.910	.47	176–181 ms
Masked surrogate	.067	.057	.883	.47	0.2 ms
TabPFN v2	.016	.013	.959	1.00	0.81–7.0 s
Proposed	.018	.015	.961	.79	48–64 ms

Reading the prediction off a network trained under random masks, the surrogate used to compute Shapley values and other removal-based explanations [21, 22], is the fastest option (0.2 ms) but its $M^{(2)}$ error, 0.057, is almost four times ours: the per-set readout is what makes a masked backbone accurate. *Retraining simple attackers per set* (linear or quadratic regression) is fast but underfits (0.096–0.144).

The one competitive method is a tabular foundation model, TabPFN v2 [23], used as an in-context attacker: it was pre-trained once on synthetic data sets and predicts the target of a new data set without training, from the training rows it receives as context. Before certification it is slightly more accurate than ours for V (0.016 against 0.018) and $M^{(2)}$ (0.013 against 0.015) and less accurate for the amplification $\Gamma^{(2)}$ (0.019 against 0.014); its advantage is concentrated on PJM, the data set with the fewest training rows, where our estimate is biased downward by 0.03–0.04, whereas on NEM it is clearly worse (error of V 0.024 against 0.010, of $\Gamma^{(2)}$ 0.032 against 0.015). The two recover comparable total marginal risk after certification: retraining the best-scanned background per field recovers 95.9% of the exact $M^{(2)}$ with TabPFN and 96.1% with ours, and three backgrounds recover 97.9% and 97.8%. Scan-certify with TabPFN recovers more critical fields at $\tau = 0.7$ (100% against 79%, the difference again coming from PJM) and fewer at $\tau = 0.5$ (88% against 91%).

The two differ in what they amortize. TabPFN amortizes learning across data sets, so every field set is a new prediction task: its training rows restricted to S form a new context that passes through an eight-member ensemble of a twelve-layer transformer with 11 million parameters, once for each target, and nothing computed for one set is reused for the next. Our estimator amortizes across the field sets of one data set: the representation is learned once, and a query costs forward passes of small networks and three linear solves, one per seed, each shared by all targets. An audit consists precisely of many such queries—thousands of sets per target and table, repeated at every re-grading—and here the difference is decisive: TabPFN takes 0.8–7.0 s per set, 13 to 110 times as long as ours; multiplying these measured costs by the query counts gives about 2.7 h versus 5 min for the NEM $K = 2$ table, and 12 h versus 54 min for the 52,360 four-field CAISO sets. Background ranking governs how much risk a small verification budget recovers. Profile accuracy also matters for reporting $M^{(0)}$ and $\Gamma^{(K)}$ across fields. The proposed estimator provides both at a per-set cost suited to repeated audits. The small positive bias of TabPFN also shows that it is at times a stronger attacker than the retrained family (Section 8); such a model is better placed as an additional member of the attacker family, where it is evaluated only on certified backgrounds, than as the scanner.

Why this estimator configuration. Table 5 summarizes three design findings. First, nonlinear features are necessary: the raw values and their squares give an $M^{(2)}$ error of 0.096, compared with 0.021 for random features. Reconstruction pre-training reduces it to 0.017 by retaining information beyond the selected targets. Second, adding an untrained feature branch reduces the error further to 0.015; the two branches supply complementary directions to the subset-specific readout. Third,

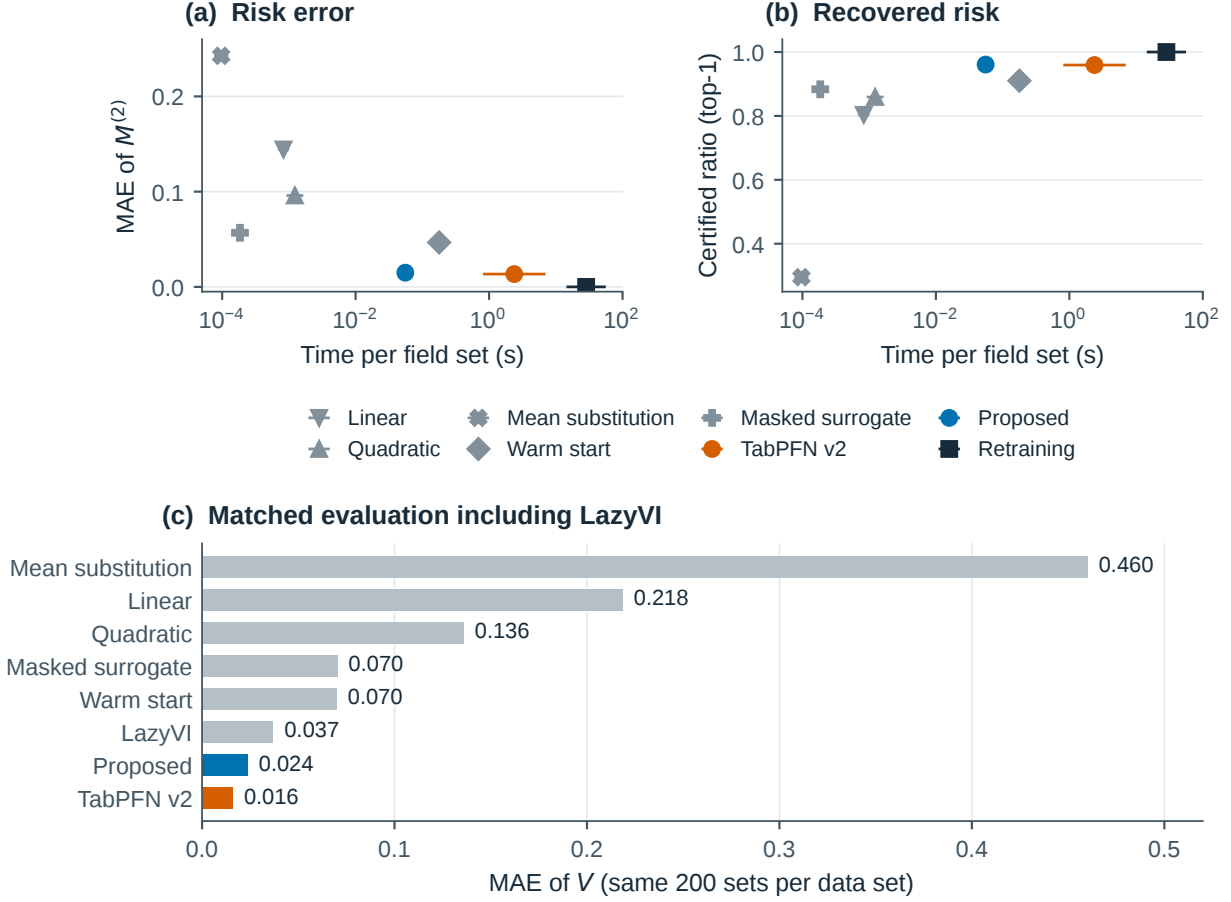


Figure 7: Comparison with other estimators over ten targets. (a) Error of $M^{(2)}$ and (b) certified top-1 ratio against the time per field set (all targets of a data set; bars: range over data sets); retraining defines the ground truth. (c) Error of V on the same 200 sets per data set for every method, including LazyVI, which was too slow to run on all sets.

Table 5: Principal estimator configurations. Errors and certified ratios are averaged over ten targets; time is the mean per-set cost for all targets of a data set. Three seeds are averaged except for the random feature row. All rows use the numerical safeguards.

Feature map	MAE V	MAE $M^{(2)}$	cert.@1	cert.@3	ms/set
Raw values and squares	.132	.096	.859	.912	1.2
Random features (one seed)	.025	.021	.958	.976	7.1
Target-only	.026	.021	.958	.976	21
Reconstruction	.020	.017	.961	.978	21
Reconstruction + random	.018	.015	.961	.978	58

stable feature scaling and prediction bounds prevent rare readout failures from dominating the estimated maximum. Without prediction clipping, the reconstruction model’s risk error rises to 0.027 and its top-1 certified ratio falls to 0.881.

The full configuration averages three independently fitted predictions and costs 58ms per set on average. We retain it as the fixed estimator because this cost is already practical for the present audit scale and it provides the most accurate overall field profiles among these configurations. Reconstruction alone costs 21 ms and achieves the same rounded certified ratios (0.961 and 0.978 for one and three backgrounds), with slightly larger risk error. It is a useful alternative when the number of rows or queries makes scanning cost the dominant concern. Appendix E reports the remaining seed, mask and representation comparisons.

Cost. All times below were measured on one otherwise idle GPU, on the same 200 random field sets per data set and with the same batch size. Retraining the attacker family sequentially for one field set takes 14–56 s (the perceptrons dominate; one network per target). Our estimator takes 48–64 ms per set, a factor of 241–906; a complete $K = 2$ table (all sets of size ≤ 3) therefore takes 1–7 min to scan, against 7–53 h of sequential retraining, and certification adds three retrained backgrounds per field and target, i.e. a few hundred sets. The time of a readout grows with the feature dimension and our estimator solves three of them, one per seed, so a one-seed estimator costs a third (16–21 ms); Tables E.10 and 4 list the time of every variant and alternative. Pre-training one reconstruction backbone takes 35–46 s; three are trained once per data set, and the untrained networks need no training. The savings compound over the recurrent workloads of an audit: re-grading after every release, redefined targets, and the sets of size four needed for $K = 3$. A backbone pre-trained by reconstructing all columns of a market can also serve every later choice of targets and candidates through the readout alone; Appendix C evaluates this re-use.

5.4. Interpreting the Profile in a Controlled Case

To interpret the measured amplification, we examine a controlled case whose inference mechanism is known in advance. We take the RTS-GMLC market of Section 4.1 and give one generating unit, 313_CC_1 (a 355 MW combined-cycle unit that is partly loaded in about 90% of hours and often sets the price), a private hourly offer markup: all its offer segments are multiplied by $(1 + \kappa_t)$, with κ_t drawn independently each hour. Offers are private information under the disclosure rules, and κ_t is the sensitive target. We clear all 8,784 hours again and assemble eleven fields: the unit’s output P_g , the LMP at its bus, the LMP at a nearby bus of the same area, the LMP at a bus behind the most frequently congested line (C6), an LMP in another area, system and area load, available wind and solar, the congestion indicator of C6, and the output of a competing unit. With eleven fields every one of the 2,047 subsets is retrained, so $M_{i \rightarrow y}^{(K)}$ is exact for every K up to the full MCI.

Prediction. When the unit is marginal, its bus price equals $(1 + \kappa_t) c_s$, where c_s is the cost of the segment in which it operates, and P_g identifies that segment. Hence $\{P_g, \lambda_g\}$ reveals κ_t while neither field does so alone; a nearby price should substitute for λ_g when the network is uncongested, a price behind a congested line should not, and loads, renewables and the congestion flag can only indicate whether the unit is marginal. We use small markups, $\kappa_t \sim U(-5\%, 5\%)$, which leave the merit order almost unchanged; larger markups ($\pm 30\%$) change the dispatch itself and are reported as a contrast.

Outcome. Table 6 and Fig. 9 confirm each prediction. Every price field has zero single-field risk and a budgeted risk of 0.10–0.15: its risk is *entirely* background amplification, and each witness contains the unit’s output. The gain of P_g itself is 0.065 alone and ranks its backgrounds exactly as the mechanism implies—own-bus price (0.215), nearby price (0.191), other-area price (0.172), remote price (0.163), competitor output (0.151), and loads, renewables and congestion (≈ 0.07). The risk also depends on the operating regime: in hours when C6 is congested the remote price loses its value as a background (gradient-boosting attacker; the joint inferability with P_g falls from 0.185 to 0.012), whereas the own-bus price becomes more informative (0.198 to 0.289). For the price fields, the budgeted risk is already close to its full-context value with small backgrounds. With $\pm 30\%$ markups the output alone already reveals much of the markup ($M^{(0)} = 0.52$), and the prices remain amplified from zero to about 0.14: the smaller the markup, the more completely it hides from single-field screening, but not from the combination.

The scores of Section 4 behave as Table G.12 predicts (Fig. G.14). Correlation, single-field retraining and pairwise inference-graph propagation assign about zero to every price. SAGE assigns 0.003 to the own-bus price, because its credit is shared among four mutually substitutable prices; LOCO likewise sees almost nothing once another price is present, and permutation importance recognizes only the one price the fitted model happens to use. $M^{(2)}$ flags all four prices and the competitor output and names the unit output as their background.

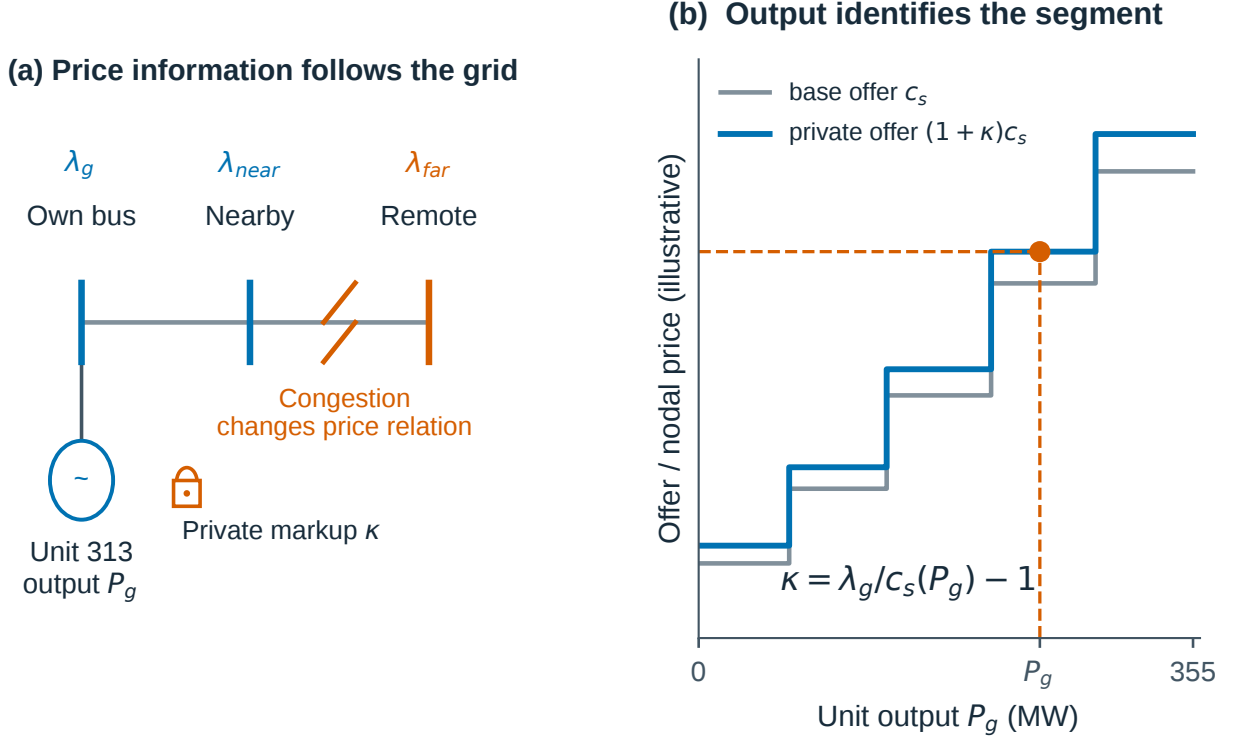


Figure 8: Physical origin of background amplification in the controlled case. Output P_g locates the active offer segment; the local price then constrains the private markup. A remote price is a weaker substitute when congestion separates price areas. The network is a schematic of the information relation, not the RTS-GMLC bus topology; the offer curve is illustrative.

Table 6: Controlled mechanism (markup $\pm 5\%$): exact field risk over all 2,047 subsets.

Field	$M^{(0)}$	$M^{(2)}$	MCI	witness at $K = 2$
Unit output P_g	0.065	0.215	0.215	own-bus LMP
Own-bus LMP	0.000	0.150	0.150	P_g
Nearby LMP (306)	0.000	0.141	0.141	$P_g + C6$
Other-area LMP (223)	0.000	0.115	0.115	$P_g + C6$
Remote LMP (303, behind C6)	0.000	0.099	0.099	P_g
Competitor output (316)	0.000	0.086	0.086	P_g
Loads, renewables, C6	0.000	0.02–0.05	≤ 0.095	–

5.5. Robustness and the Background Budget

Attacker seeds and data splits. Ground truth is itself an estimate: attackers are trained with random initializations and early-stopping splits, and V_y is measured on one random test split. We repeated the complete three-attacker retraining with two further seeds (RTS-GMLC, three targets; PJM load) and on two further random 70/30 splits (RTS-GMLC). Across these reruns the mean absolute change in V_y is 0.001–0.008, the Spearman correlation of $M^{(2)}$ across fields with the main run is 0.983–0.999, the set of τ -critical fields is identical at $\tau = 0.5$ (Jaccard 1.0 in every case) and agrees with Jaccard 0.93–1.00 at $\tau = 0.7$, and the number of combination-critical fields changes by at most one or two fields (Fig. 10c). The conclusions of Sections 5.1–5.2 are therefore not artefacts of a particular seed or split.

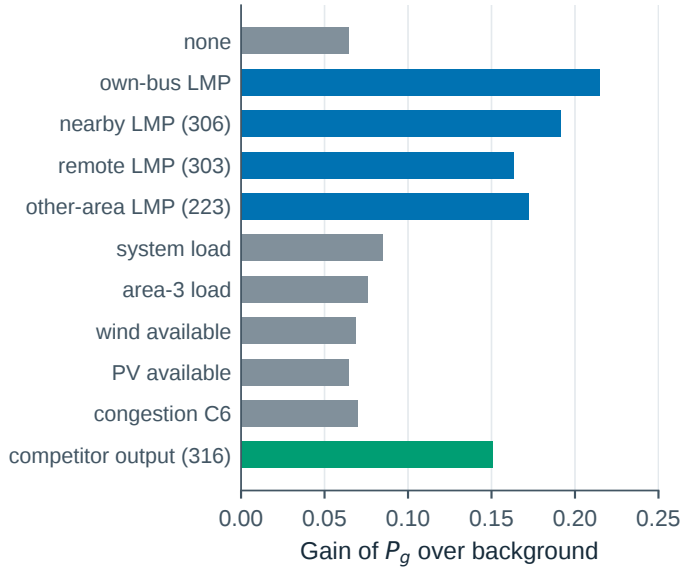
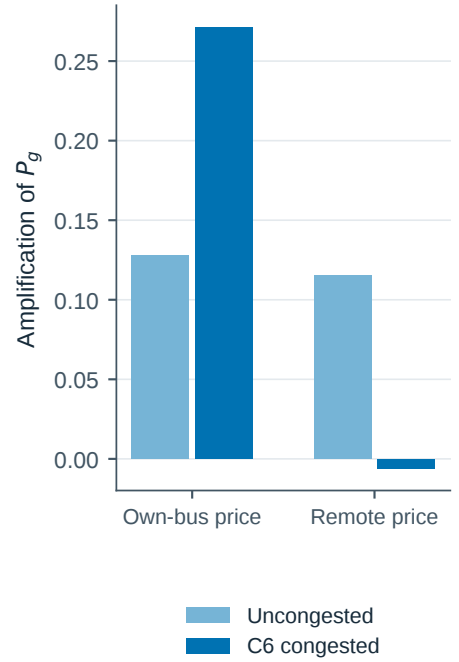
(a) Context changes the gain**(b) Congestion matters**

Figure 9: Controlled mechanism: private offer markup of unit 313_CC_1 ($\pm 5\%$). (a) Gain of the unit output P_g over each single background. (b) Amplification of P_g by the own-bus and the remote price in uncongested hours and in hours when line C6 is congested.

Threshold. Fig. 10(d) sweeps τ from 0.3 to 0.9 over all ten targets. At every threshold the share of fields that are safe alone but critical with at most two background fields exceeds the share of fields that are unsafe alone (e.g. 43% versus 10% at $\tau = 0.7$). The gap between the two grading philosophies is thus not tied to the choice of τ , although its size is (Section 6).

Choice of K . Exhaustive retraining of all sets of four fields on every data set (for NEM an additional 27,405 sets) gives the exact $M^{(3)}$. Measured by the *marginal* risk, $K = 2$ is close to saturation: $M^{(2)}/M^{(3)}$, the ratio of mean field risk, lies between 0.958 and 1.000 for nine targets and is 0.813 for NEM TUMUT3, a hydro unit whose dispatch depends on inter-regional exchange (Fig. 10a). The first background field brings most of the amplification ($M^{(1)}/M^{(2)}$ between 0.73 and 0.97).

Threshold crossings, however, do not saturate as quickly, as expected for budget-dependent criticality: a small increase of $M_{i \rightarrow y}^{(K)}$ can move a field across τ . At $\tau = 0.7$ several targets have fields that become critical only with a third background field: PJM net interchange (3 critical fields at $K = 2$, 21 at $K = 3$), NEM PPCCGT (10 and 24) and PJM load (14 and 22) (Fig. 10b). At $\tau = 0.5$ all fields are already critical at $K = 2$ except for TUMUT3 (22 versus 30). Taken together, these results show that small backgrounds recover most of the measured marginal risk, while the number of critical fields remains sensitive to the assumed budget. A stronger background model is evaluated by increasing K and reporting the corresponding criticality results. Evaluating all sets of size four is the workload for which the amortized scan is needed: for CAISO it comprises 52,360 sets per target, against 7,175 for $K = 2$.

6. Application to Data-Release Decisions

The field-level risk is a measurement; what an operator does with it depends on a threshold τ and on policy. We illustrate one downstream use and state its limits.

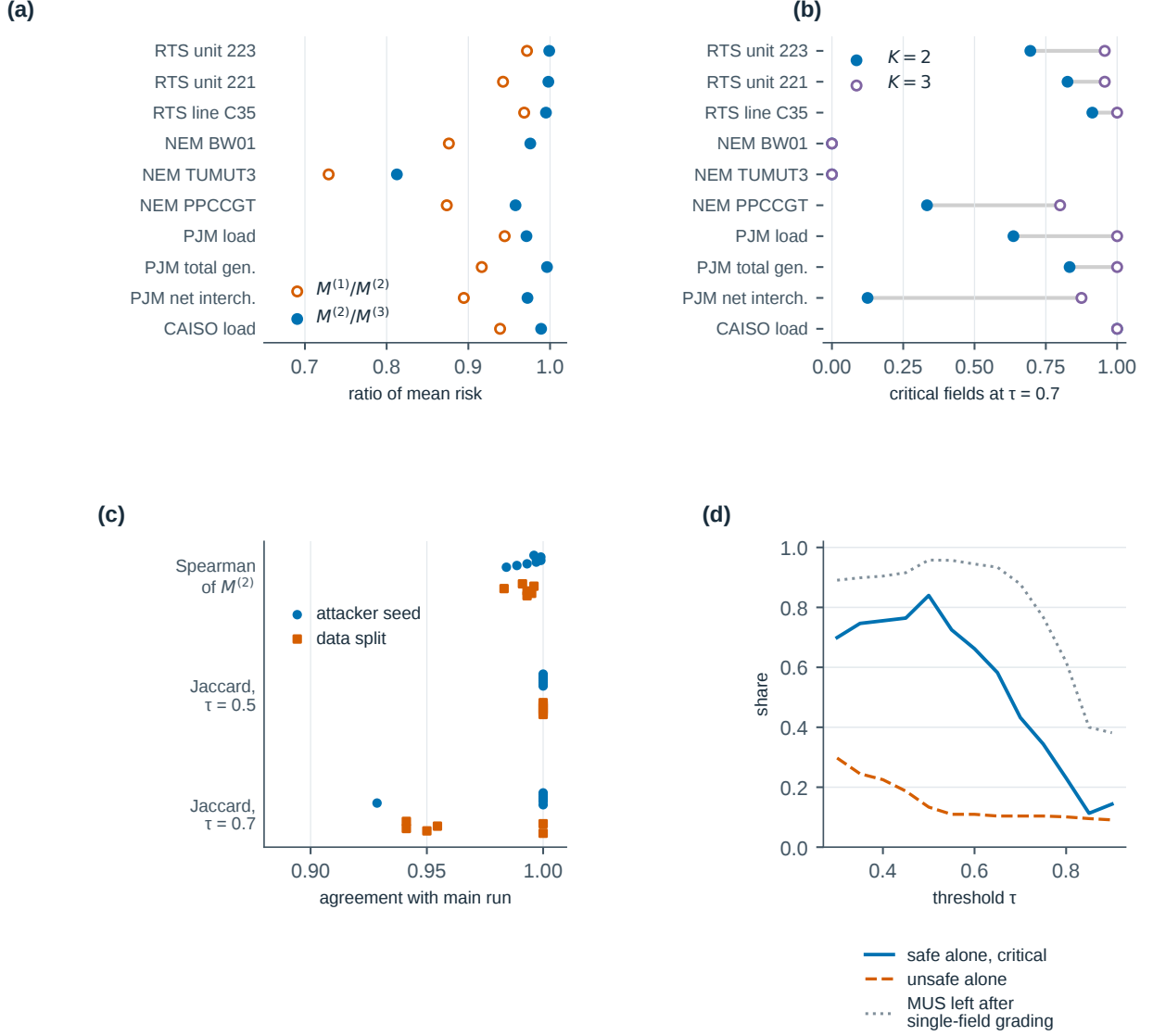


Figure 10: Robustness. (a) Saturation of mean field risk in the background budget. (b) Share of τ -critical fields at $\tau = 0.7$ with $K = 2$ (filled) and $K = 3$ (open). (c) Agreement of reruns with different attacker seeds or data splits with the main run. (d) Threshold sweep over ten targets: fields that are safe alone but critical (solid), fields unsafe alone (dashed), and minimal unsafe sets left after single-field grading (dotted).

Critical fields. Given τ , Theorem 2 turns the risk table into a flag: field i is critical if some background of at most K fields is safe on its own and becomes unsafe when i is added. Over the ten targets the single-field rule $V_y(\{i\}) > \tau$ recovers only 13% ($\tau = 0.5$) and 17% ($\tau = 0.7$) of the critical fields. Scanning with the amortized attacker and certifying at most three candidate backgrounds per field by retraining recovers 76–91% of them with no false positives (Table 7).

Withholding. The minimal unsafe sets identify which fields to withhold. Withholding every field that is unsafe alone leaves 98% ($\tau = 0.5$) and 96% ($\tau = 0.7$) of the minimal unsafe sets of size ≤ 3 intact. Two strategies that choose fields from estimates alone, without certification—a minimum hitting set of the estimated minimal unsafe sets, and a greedy rule that withholds the field with the largest $\hat{M}^{(2)}$ and recomputes $\hat{M}^{(2)}$ over the remaining fields—leave 0.3–1.6% intact while withholding at most 1.9 fields more than the ground-truth optimum.

Limits. These numbers depend on the threshold, which is a policy choice: at $\tau = 0.5$ almost every field is critical in our data, at $\tau = 0.9$ very few are, and the share of dangerous combinations that

Table 7: Decisions over the ten targets. Top: recall and precision of τ -critical fields. Bottom: fields withheld (optimum in parentheses) and share of true minimal unsafe sets left unbroken. Selection uses estimates only; retrained values are used for certification and evaluation. δ : conservative margin of the scan.

Flagging rule	$\tau = 0.5$		$\tau = 0.7$	
	recall	precision	recall	precision
Single-field rule $V(\{i\}) > \tau$	0.13	1.00	0.17	1.00
Scan on $\hat{V}$, $\delta = 0.05$	0.92	0.99	0.84	0.76
Scan + certify, $\delta = 0$	0.91	1.00	0.79	1.00
Scan + certify, $\delta = 0.05$	0.76	1.00	0.80	1.00
Withholding strategy	withheld	left	withheld	left
Single-field grading	3.3 (9.4)	98.2%	2.5 (5.1)	95.5%
Hitting set of MUS on $\hat{V}$, $\delta = 0.05$	10.0 (9.4)	1.5%	5.4 (5.1)	1.6%
Adaptive $\hat{M}^{(2)}$ greedy, $\delta = 0.05$	11.3 (9.4)	0.3%	6.3 (5.1)	1.3%

single-field grading leaves intact varies from 96% to 38% across $\tau \in [0.5, 0.9]$. They also depend on K : minimal unsafe sets of size four exist that a withholding set computed at $K = 2$ does not break, and breaking them requires on average 11.3 instead of 9.4 withheld fields at $\tau = 0.5$ and 6.3 instead of 5.1 at $\tau = 0.7$ (Section 5.5). A release decision on a concrete set A can also be checked directly by evaluating $V_y(A)$. Withholding is therefore an application of the risk profile rather than its purpose: it shows that the per-field quantity carries enough structure—witnesses and minimal unsafe sets—to act on.

7. Related Work

Inference and aggregation in data release. That individually innocuous data items can jointly reveal protected information is an old observation in database security, where it is known as the inference and aggregation problem [1, 2]; the privacy literature supplies the best-known illustration, three demographic attributes that jointly single out most individuals [24]. In power systems, public market data have been shown to expose non-public physical information, for example grid topology from locational marginal prices [5]. Recent work studies reconstruction through attribute correlations in multi-attribute releases [3]. This is closely related to the motivation for measuring conditional contributions, while our audit derives a field profile from the inferability of admissible sets. Privacy mechanisms for metering data [25, 26] and formal models such as differential privacy [27] or k -anonymity [28] bound what can be learned about an individual record after a release mechanism is applied. They answer a different question from the one faced by a market operator, who must decide which aggregate operational *fields* may be published at all.

Data sensitivity assessment under inference leakage. Data classification and grading is performed field by field in practice, and the Chinese national standard explicitly asks that aggregation and processing effects be considered [29], without prescribing how. The closest work to ours is IGNN [6], which introduced data inference leakage into sensitivity assessment for cyber-physical power grids: fields are nodes of an inference graph whose edges encode physical or statistical inference relations, and sensitivity is propagated along these edges by a graph attention network from an intrinsic (expert-assigned) score. This established that the risk of a field depends on its inference relations to sensitive data. Two questions remain open in this line of work. First, the risk score is calibrated against expert labels, while its ultimate validation is an attack experiment in which a model is trained on a group of fields; the attack outcome itself is not the definition. Second, an edge connects two fields, whereas inference frequently requires several fields jointly, with none of them informative

alone. We take the attack outcome on a field *set* as the definition of risk and derive the field-level quantity from it.

Feature importance. Turning set-level values into per-feature scores is the subject of feature attribution. Shapley-based measures [7, 18, 30, 31] average a feature’s marginal contribution over all contexts, which is the right answer for fair allocation but dilutes the credit of redundant features [9, 32]. Leave-one-covariate-out (LOCO) [8] and permutation importance [17, 33, 34] evaluate a single context, the full feature set, so a feature with a near-substitute receives almost no importance. The Marginal Contribution Index (MCI) [9] takes the *maximum* marginal contribution over contexts, is characterized axiomatically, and is immune to dilution. Security reasoning favours the worst case for the same reason quantitative information flow replaced Shannon entropy by min-entropy and gain functions [35, 36]: an average over contexts is not a guarantee. Our field-level risk is a budget-constrained MCI evaluated relative to already-public data.

Fast estimation of many refitted models. Computing the performance of a model refitted on many feature subsets is expensive, which has motivated lazy [19] and warm-started [20] estimators of variable importance, surrogate models trained under random masks for removal-based explanations [21, 22, 37], conditional generative models trained under random masks [10, 11, 38, 39], and tabular foundation models that predict from any set of columns in context [23]. The last amortize learning across data sets: each query passes the training rows of the requested columns through a large pre-trained transformer. Our estimator amortizes across the field sets of one data set instead, so that the per-set work reduces to a linear solve on shared features. Closed-form ridge heads on frozen features are the standard fast-adaptation device in meta-learning [40, 41], and the benefit of sharing a representation across tasks is well understood [42–44]. We combine these ingredients into an estimator of set-level inferability and, more importantly, use it inside a scan–certify loop so that decisions rest on retrained attackers rather than on the estimate.

Efficiency evaluated against the security task. Practical security methods must account for the cost of producing useful evidence. For example, LPASS uses linear probes to estimate the performance of compressed models for vulnerability detection and evaluates both security-task performance and computational savings [45]. Our application has a different workload: thousands of related field subsets within one data set. The shared estimator is evaluated against retrained inferability and downstream background recovery, rather than against prediction accuracy alone.

8. Discussion

8.1. Security meaning of the profile

The proposed measurement makes background information part of the field-level result. A large single-field score identifies direct exposure. Large amplification identifies exposure that depends on an additional combination, and the witness backgrounds specify that combination. These distinctions remain visible even when two fields have similar overall risk. They are useful for a security analyst who needs to explain why a release warrants review.

The maximum contextual gain also serves a different purpose from average attribution. In the evaluated data, the median average-to-maximum gain ratio is 0.32, and many high-gain backgrounds form small families around a shared partner. The profile retains these backgrounds instead of dispersing their contribution across an average. The safety-margin result formalizes the worst-case gain, while criticality identifies threshold crossings under the selected budget.

8.2. What should be measured in an efficient audit

Profile accuracy and verification coverage capture different aspects of performance. Set-value and marginal-risk errors determine how faithfully the scanner describes a field. Background ranking determines which evidence can be recovered with a small retraining budget. Critical-field recall

additionally depends on the threshold, so comparable recovered marginal risk need not produce the same set of flags.

The foundation-model comparison illustrates this distinction. TabPFN v2 and the proposed estimator recover almost the same total marginal risk after retraining, but TabPFN has higher critical-field recall at $\tau = 0.7$. The proposed estimator is more accurate for amplification and is substantially cheaper to apply across many subsets. Reporting these results separately makes the operating trade-off clear without reducing audit quality to one metric.

The fixed reconstruction-plus-random configuration also reflects both purposes. Its millisecond scan cost is practical for the present data scale, and its modest improvement in profile accuracy is useful when reporting risk across all fields. The reconstruction-only alternative reduces mean cost from 58 to 21 ms with the same rounded certified ratios, making it an appropriate option for larger query or row counts. This is a choice of estimator configuration within one audit framework.

8.3. Applicability and extensions

The audit assumes labelled historical information for supervised inference and evaluates a specified attacker family. Changes in data distribution, public baseline or attacker capability require the reference and estimated profiles to be updated. Random train/test splits and the selected operational targets define the empirical scope of the present study. The background budget is part of the result: increasing it reveals additional threshold crossings even where mean marginal risk is close to saturation.

A useful extension is to enrich background proposals with physical relations or inference graphs. Another is to add stronger predictors to the verification family. The all-column experiment suggests that feature reuse can also reduce the cost of revising target definitions. These directions preserve the separation between a field’s estimated profile and the combinations verified by dedicated retraining.

9. Conclusion

We presented an audit for background-dependent inference leakage that derives field-level risk from the inferability of data sets. Budgeted marginal risk records the largest admissible gain, background amplification identifies the part missed by single-field assessment, and witness backgrounds explain the contributing combinations. Shared reconstruction and random features with subset-specific readouts make these profiles efficient to estimate; retraining selected backgrounds verifies the evidence.

Across four power-system data sets, the estimator achieves a marginal-risk error of 0.015 and recovers 97.8% of total marginal risk after verifying three backgrounds per field. Critical-field recall is 76–91% without false positives in the evaluated settings, compared with 13–17% for single-field screening. The combined definition, efficient measurement and verification procedure provide a practical basis for auditing inference risk before data release.

Data and code availability

RTS-GMLC [13] and the AEMO NEMWEB archive [14] are public; PJM and CAISO data were obtained from the operators’ public portals. Field-role tables, dispatch simulation, reference retraining, estimator and analysis code will be made available upon publication.

Appendix A. Further Properties of Budgeted Marginal Risk

Proposition 3 (Ladder and candidate-set monotonicity). (a) $M_{i \rightarrow y}^{(0)} \leq M_{i \rightarrow y}^{(1)} \leq \dots \leq M_{i \rightarrow y}^{(|H|-1)}$; (b) enlarging the candidate set from H to $H' \supseteq H$ cannot decrease $M_{i \rightarrow y}^{(K)}$; (c) if f' duplicates f , i.e. $V_y(B \cup T \cup \{f'\}) = V_y(B \cup T \cup \{f\}) = V_y(B \cup T \cup \{f, f'\})$ for all T , deleting f' leaves $M_{i \rightarrow y}^{(K)}$ unchanged for every $i \neq f, f'$.

Proof. (a) and (b) are maxima over nested feasible sets. (c) If a witness T of i contains f' , replace f' by f (if $f \notin T$) or drop it (if $f \in T$); the value is unchanged and the budget is respected, so the maximum does not decrease after deleting f' , and by (b) it does not increase. $\square$

Theorem 3 (Truncated super-efficiency). *For $A \subseteq H$ with $|A| \leq K + 1$, $V_y(B \cup A) - V_y(B) \leq \sum_{i \in A} M_{i \rightarrow y}^{(K)}$. More tightly, with $U^{(K)}(\emptyset) = 0$ and $U^{(K)}(A) = \min_{j \in A} [U^{(K)}(A \setminus \{j\}) + M_{j \rightarrow y}^{(\min(|A|-1, K))}]$, $V_y(B \cup A) - V_y(B) \leq U^{(K)}(A) \leq \sum_{i \in A} M_{i \rightarrow y}^{(K)}$.*

Proof. Order $A = \{i_1, \dots, i_m\}$ and telescope $v_B(A) = \sum_t [v_B(A_t) - v_B(A_{t-1})]$; the t -th term is a marginal of i_t over a background of size $t - 1 \leq K$, hence at most $M_{i_t \rightarrow y}^{(K)}$. The staged bound follows by induction on $|A|$ and Proposition 3(a). $\square$

For $K \geq |H| - 1$ this is the super-efficiency of MCI [9]; with a finite budget it is stronger because $M_{i \rightarrow y}^{(K)} \leq \text{MCI}$. It yields a release rule “accept A if $V_y(B) + U^{(K)}(A) \leq \tau$ ” that never accepts an unsafe set of size $\leq K + 1$. We do not use this rule in the main text because it is conservative for larger release sets; withholding a minimum hitting set of the minimal unsafe sets (Section 6) addresses the same need directly.

Proposition 4 (Irreducible synergy is bounded by field risk). *With $\text{syn}^B(S) = V_y(B \cup S) - \max_{T \subsetneq S} V_y(B \cup T)$ and $|S| \geq 2$, $\text{syn}^B(S) \leq \min_{j \in S} M_{j \rightarrow y}^{(|S|-1)}$.*

Proof. $\max_{T \subsetneq S} V_y(B \cup T) \geq V_y(B \cup S \setminus \{j\})$ for every j , hence $\text{syn}^B(S) \leq \Delta_j(B \cup S \setminus \{j\}) \leq M_{j \rightarrow y}^{(|S|-1)}$. $\square$

Remark 1 (Amplification is weaker than synergy). $M_{i \rightarrow y}^{(K)} = M_{i \rightarrow y}^{(0)}$ iff $\Delta_i(B \cup T) \leq \Delta_i(B)$ for all $|T| \leq K$. Submodularity implies no amplification but not conversely: with $V_y(\{1\}) = 0.8$, $V_y(\{2\}) = 0.6$, $V_y(\{3\}) = 0.8$, all pairs 0.8 and $V_y(\{1, 2, 3\}) = 1$, every field has $M^{(2)} = M^{(0)}$, yet $\Delta_1(\{3\}) = 0 < \Delta_1(\{2, 3\}) = 0.2$.

Appendix B. Implementation Details

Attackers. Single- and multi-target perceptrons: two hidden layers of 128 ReLU units, dropout 0.15, Adam (learning rate 10^{-3} , weight decay 5×10^{-4}), batch 128, at most 300 epochs, patience 60, epoch chosen per (set, target) on validation. Many attackers are trained in parallel on one GPU as independent networks whose inputs are masked to their own field set; masked columns receive no gradient, so each network is functionally a network trained on its set alone. Gradient boosting: histogram gradient boosting with (learning rate, leaves) $\in \{(0.05, 15), (0.1, 31)\}$, 300 iterations with internal early stopping; the configuration is chosen on validation. The family member is chosen per (set, target) on validation. Across data sets gradient boosting is selected for 35–76% of (set, target) pairs (RTS-GMLC 76%, NEM 35%, PJM 67–69%, CAISO 52%), so the family maximum matters.

Estimator. Backbone: three hidden layers of 256 ReLU units with dropout 0.1, input $[x \odot m, m]$, Adam (10^{-3} , weight decay 5×10^{-4}), batch 256, at most 400 epochs, patience 60 on the target loss under eight fixed validation masks. The output layer predicts the targets and all candidate columns; the reconstruction loss is averaged over hidden columns only. The untrained networks share the architecture and are initialized with the default PyTorch initialization under seeds 0, 1, 2; seed r of the estimator pairs the reconstruction backbone and the untrained network of seed r . Ridge penalties $\lambda \in \{10^{-3}, \dots, 10^2\}$. The readout computes $Z^\top Z$, $Z^\top y$ and the column sums once on all training rows and obtains those of each cross-validation fold by subtracting the held-out rows; the six penalties are solved as one batched linear system. This changes only the order of computation (estimates are identical to a direct implementation) and makes a readout about 2.8 times faster; all times reported use it. Queries are processed in batches of eight field sets.

Estimators compared. All use the same standardized data, training rows and random split as the ground truth, and default settings. *Full model:* a multi-output perceptron with two hidden layers of 128 ReLU units (dropout 0.15), trained on all candidate fields (Adam, 10^{-3} , weight decay 5×10^{-4} , batch 128, at most 300 epochs, early stopping with patience 60 on validation rows), three seeds. *Mean substitution:* the removed fields are set to their training mean (zero after standardization) and the three full models’ predictions are averaged. *LazyVI:* the full model of seed 0 is linearized in its parameters at the inputs with removed fields set to zero; the residual $y - f(\tilde{x}; \theta_0)$ is fitted by kernel ridge regression with the neural tangent kernel $JJ^\top$, one output at a time, with the penalty chosen by exact leave-one-out cross-validation from $\{10^{-3}, \dots, 10^3\}$ times the mean eigenvalue; the kernel is formed and decomposed in double precision (single precision was ten times faster but visibly less accurate). *Warm start:* from the full model of seed 0, at most 100 Adam steps (batch 256) on the inputs with removed fields set to zero, validation loss checked every ten steps and the best state kept. *Masked surrogate:* the target-only masked backbones’ own outputs, averaged over three seeds. *TabPFN v2:* package `tabPFN` 2.2.1 with its default ensemble of eight members, one regressor per target, fitted in context on the training rows of the columns in S . Times are measured on one otherwise idle GPU, for all targets of a data set; the full models take 11–18 s each to train.

Inference-graph baseline. Following the structure of [6] without its expert-assigned intrinsic scores: nodes are the candidate fields and the target; the weight of edge $i \rightarrow j$ is the validation R^2 of predicting field j from field i alone by gradient boosting; the score of i is $\sum_{h=1}^3 0.5^{h-1} (W^h)_{i,y}$. The first term equals single-field predictability; longer paths add inference chains through other fields.

SAGE. Marginal imputation with 32 background rows, 512 evaluation rows from the validation split and 24 random permutations, on a gradient-boosting model fitted to all candidates.

Table B.8: Field roles and rule basis (Chinese electricity-market disclosure rules [4]; appendix item numbers in parentheses).

Data set	Candidate fields (released before the target)	Targets
RTS-GMLC	zonal load forecasts (3; 6.41); wind, PV, rooftop PV and hydro forecasts (4; 6.42); regulation up/down and spinning reserve requirements (3; 6.46); system energy price and nodal prices at 9 buses (10; 6.52); congestion indicators of lines C6, C29, CA-1 (3; 6.54)	outputs of units 223_STEAM_3 and 221_CC_1 (specific information, Art. 17(3)); hourly flow on line C35 (only daily averages of important lines are listed, 6.66)
NEM 2024	regional demand (5), available capacity (5), semi-scheduled wind and solar forecasts (9), regional prices (5), interconnector flows (6)	actual outputs of units BW01, TUMUT3, PPCCGT (specific information)
PJM	regulation market prices and volumes (5); day-ahead reserve prices, requirements, self-schedules and cleared volumes (9); day-ahead system energy price, LMP, congestion and loss components (4); real-time LMP, congestion and loss components (3); scheduled interchange (1); for total generation and net interchange also the system load forecasts (2)	metered load (6.61), total generation (6.57), net actual interchange (6.65), all released next day
CAISO	day-ahead load forecasts of three sub-areas (3); day-ahead wind and solar forecasts (6); day-ahead and 15-min real-time prices and components (11); day-ahead schedules of sub-area load, total generation, imports and exports (6); ancillary-service requirements incl. regulation mileage (5) and procured volumes (4)	actual CAISO load (6.61), released next day

Appendix C. One Backbone for All Target Definitions

In practice the auditor’s target list and candidate pool change—a field is reclassified, a new target is added. With masked reconstruction over all 53 columns of a market, one backbone serves

every such definition through the readout alone. Table C.9 compares it with backbones supervised on each group’s own targets. The all-column backbone is more accurate for V (0.040 versus 0.049) and slightly more accurate for $M^{(2)}$ (0.026 versus 0.029), and its certified top-1 ratio rises from 0.906 to 0.944.

The mechanism is visible in the representation. A backbone that serves one or two targets leaves a quarter of its hidden units inactive (24%) and has an effective dimension of 1.45; one that must reconstruct every column keeps 92% of units active and has an effective dimension of 2.44. It has to encode the latent operating state—load level, renewable output, congestion—that drives all grid quantities, as argued for multi-task representation learning [42–44] (Fig. C.11). Why V improves more than M follows from (8): 89–98% of the V error is common-mode, and a better representation mostly shrinks this common part, which cancels in the marginals.

The last row of Table C.9 tests a target never seen in pre-training: the target column is deleted from the data before pre-training. Set values and marginals are estimated as well as with the all-column backbone, but $\hat{M}^{(2)}$ is inflated (bias +0.028), because the maximum over many marginals picks up the few that are overestimated; the certified top-1 ratio drops to 0.75. Retraining three backgrounds per field restores it to 0.966. Re-use for unseen targets therefore requires certification, which the scan–certify pipeline provides.

Table C.9: Backbone scope on the four PJM/CAISO targets (same readout). Inactive: hidden units with zero variance on the training rows; eff. dim: participation ratio of the feature covariance; common-mode: $|\bar{e}|/|e|$ of the V error e .

Backbone	inactive	eff. dim	MAE V	common-mode	MAE Δ	MAE $M^{(2)}$	bias $M^{(2)}$	cert.@1	cert.@3
Per-group (targets supervised)	24.2%	1.45	0.049	98%	0.028	0.029	-0.025	0.906	0.963
All-column (masked reconstruction)	7.7%	2.44	0.040	94%	0.024	0.026	-0.021	0.944	0.976
All-column, target held out	7.5%	2.50	0.039	94%	0.024	0.056	+0.028	0.750	0.966

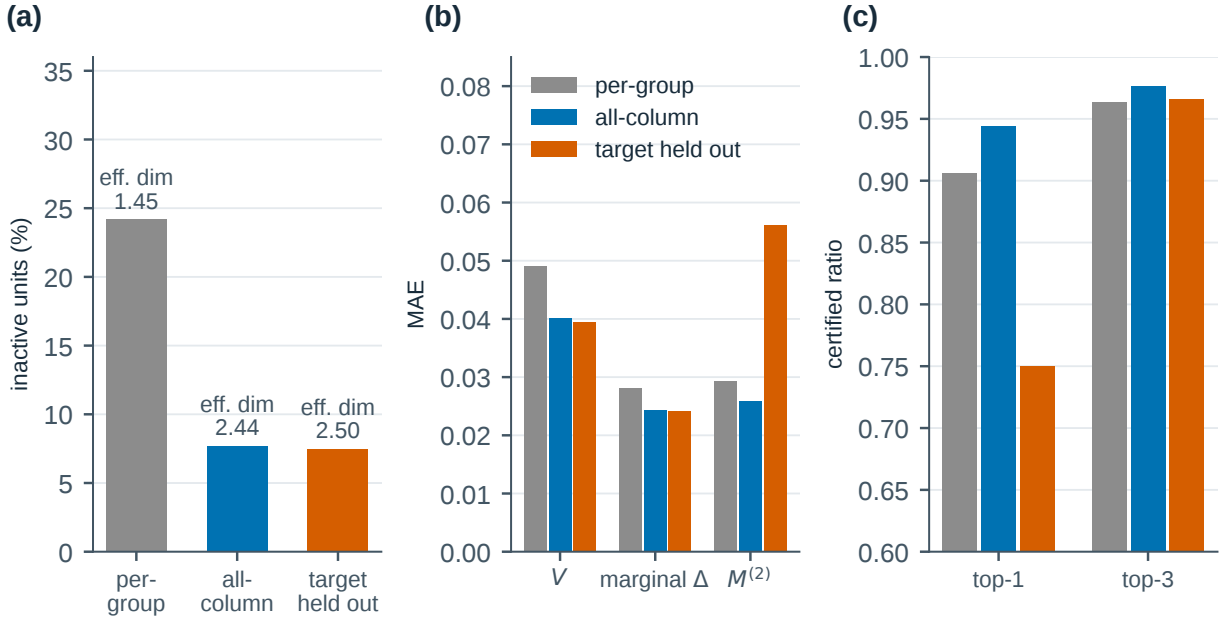


Figure C.11: Backbone scope on the four PJM/CAISO targets (per-set readout on each backbone alone, three seeds): (a) inactive hidden units and effective dimension, (b) error of V , of marginals and of $M^{(2)}$, (c) certified ratios.

Appendix D. A Synthetic Example

Fig. D.12 uses synthetic data with a known structure, $Y = aAB + bP + \sigma\varepsilon$, where A and B act only jointly, C_1 and C_2 are two noisy copies of P , D is a weak copy of P and N is noise; all 63 sets are retrained. Correlation, single-field retraining and pairwise inference-graph propagation give A and B a score of zero although $\{A, B\}$ reaches 0.54. LOCO gives C_1 and C_2 almost zero because each substitutes for the other, and LOCO, permutation importance and SAGE all give D zero because the full model relies on C_1, C_2 instead—yet $\{A, B, D\}$ crosses $\tau = 0.7$ while $\{A, B\}$ does not, so D is critical. SAGE splits credit within both pairs. $M^{(2)}$ assigns A, B, C_1, C_2 their full worst-case contribution and gives D a non-zero risk with the witness $\{A, B\}$. This is the behaviour predicted by Table G.12: averages dilute, single contexts miss substitutes, and pairwise edges miss joint inference.

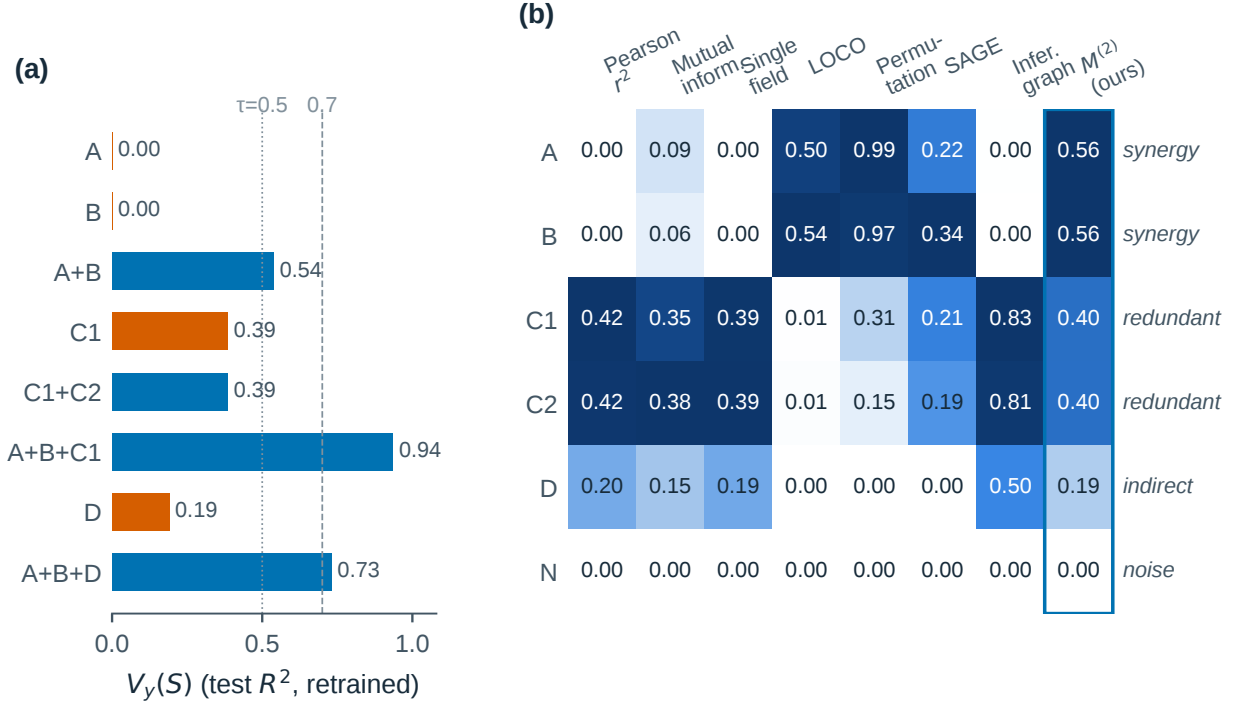


Figure D.12: Synthetic mechanism example. (a) Retrained inferability of selected sets. (b) Scores of eight methods (colour normalized per column); the last column is $M^{(2)}$.

Appendix E. Estimator Design Diagnostics

The comparisons below examine representation, seed aggregation and numerical stability under the existing experimental protocol. They supplement the principal configurations in Table 5.

Representation. Target-only masked pre-training yields an effective feature dimension of 1.60, compared with 3.30 for an untrained network. Restricting masks to one to four visible fields gives 1.65. Reconstruction raises the dimension to 2.75. At two retained principal components, trained features have about half the set-value error of random features, but the remaining dimensions supply much of the small-set readout’s accuracy. For 16-field queries, random features have error 0.074, compared with 0.027 for target-only and 0.023 for reconstruction features; this larger-set diagnostic uses a gradient-boosting reference and is separate from the exhaustive main audit.

Seeds and numerical stability. Averaging three predictions reduces the full estimator’s set-value error from 0.020 to 0.018. Concatenating the features of three target-only seeds into one readout

costs 41 ms, compared with 21 ms for averaging their separate predictions, and yields a lower top-1 certified ratio on the six targets used in that comparison (0.916 versus 0.953). Rare unstable predictions disproportionately affect the maximum marginal gain: without prediction clipping, the reconstruction configuration has risk error 0.027 and certified ratio 0.881, despite its 0.020 set-value error. The safeguards in Section 3 address this failure. Uniform visible-count and Bernoulli masks give similar risk errors (0.023 and 0.024); eliminating masks raises the error to 0.036.

Table E.10: Design choices of the estimator, averaged over the ten targets and all sets of size ≤ 3 . The untrained network is paired with the per-set readout. Cert.@ k : certified $L^{(2)}$ over exact $M^{(2)}$ when k backgrounds per field are retrained. ms: time per field set on one idle GPU (same 200 sets per data set, batch 8, mean over data sets); – : same cost as the corresponding reference row. [†]Evaluated on the six RTS-GMLC and NEM targets only; on the same targets three averaged target-only seeds give 0.011, 0.016, 0.973, 0.953 and 0.974.

Configuration	MAE V	MAE $M^{(2)}$	Spearman	cert.@1	cert.@3	ms
<i>Feature map (three seeds averaged, all safeguards)</i>						
None: readout on $[x_S, x_S^2]$	0.132	0.096	0.854	0.859	0.912	1.2
Untrained network (one seed)	0.025	0.021	0.963	0.958	0.976	7.1
Target-only pre-training	0.026	0.021	0.960	0.958	0.976	21
Reconstruction pre-training	0.020	0.017	0.963	0.961	0.978	21
Reconstruction $\oplus$ untrained (ours)	0.018	0.015	0.968	0.961	0.978	58
<i>Use of seeds</i>						
Ours, one seed (mean over three single-seed estimators)	0.020	0.016	0.965	0.960	0.977	19
Target-only, one seed	0.030	0.023	0.953	0.936	0.967	7.1
Target-only, three seeds, predictions averaged	0.026	0.021	0.957	0.934	0.970	21
Target-only, three seeds, features concatenated in one readout [†]	0.011	0.013	0.958	0.916	0.962	41
<i>Numerical safeguards</i>						
Reconstruction, three seeds, without prediction clip	0.020	0.027	0.921	0.881	0.974	21
Target-only, one seed, without any safeguard	0.031	0.036	0.904	0.825	0.946	–
<i>Pre-training masks and raw features (target-only, one seed)</i>						
No masks (always fully visible)	0.032	0.036	0.928	0.898	0.963	–
Bernoulli(0.5) masks	0.030	0.024	0.958	0.932	0.969	–
Readout on ϕ only, without $[x_S, x_S^2]$	0.030	0.025	0.950	0.927	0.964	–

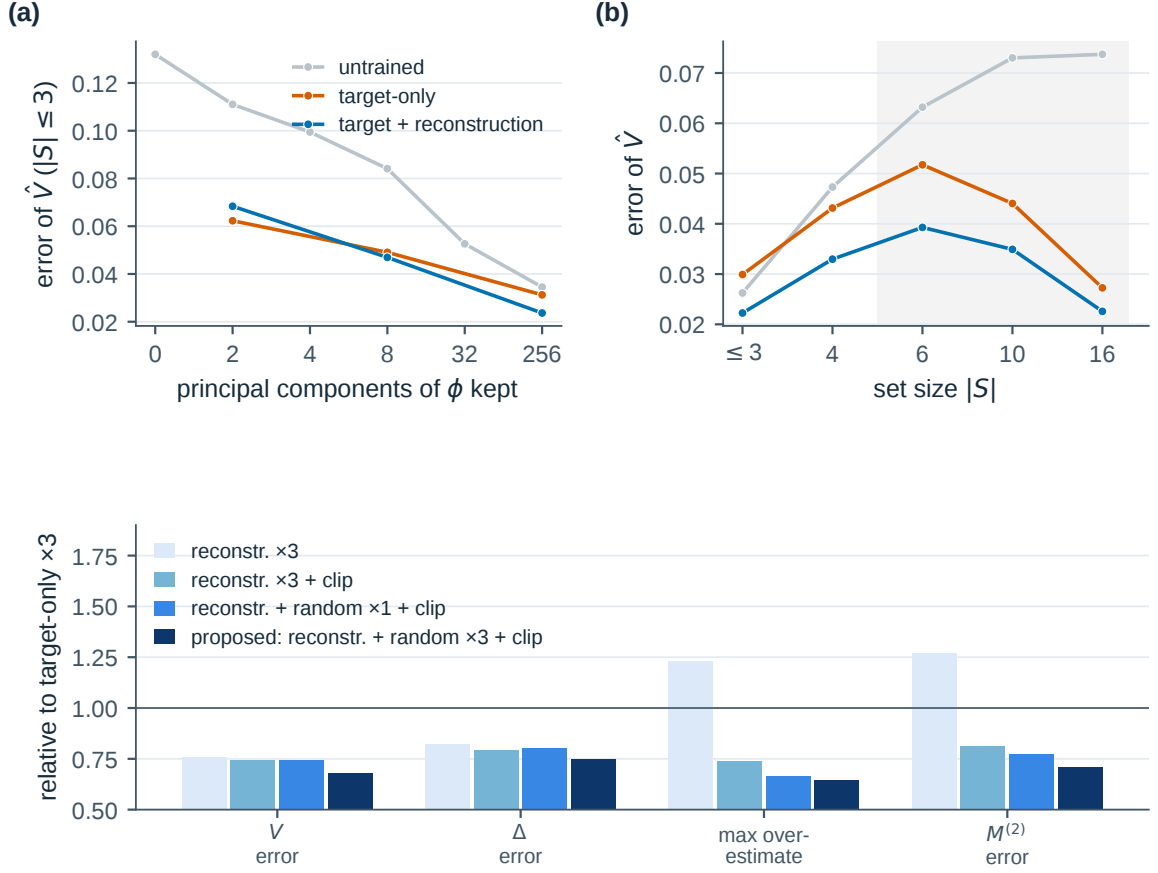


Figure E.13: Representation and stability diagnostics. (a) Set-value error when only leading principal components are retained. (b) Error with increasing set size; shaded queries use a gradient-boosting reference. (c) Effects of clipping and the random feature branch on set-value, marginal and maximum-risk errors, relative to the target-only three-seed configuration.

Appendix F. Additional Baseline Metrics

Table F.11: Additional baseline metrics. MAE V_{200} uses the same 200 sampled sets for every method. Other columns use the complete small-set audit. Dashes indicate metrics not evaluated.

Estimator	MAE V_{200}	MAE $\Gamma^{(2)}$	Spearman	Recall .5	Recall .7
Linear	.218	.050	.737	.24	.16
Quadratic	.136	.049	.854	.31	.27
Mean substitution	.460	.111	.534	.03	.00
LazyVI	.037	—	—	—	—
Warm start	.070	.028	.932	.67	.47
Masked surrogate	.070	.034	.930	.55	.47
TabPFN v2	.016	.019	.979	.88	1.00
Proposed	.024	.014	.968	.91	.79

Appendix G. Field-Score Requirements and Controlled Scores

Table G.12: Requirements for a field-level inference-risk quantity. ✓: satisfied; ○: partially; −: not satisfied.

Method	checkable by attack	joint inference	worst case	no dilution	bounded K	public baseline B
Correlation / mutual information	−	−	−	✓	−	−
Single-field retraining $M^{(0)}$	✓	−	−	✓	−	○
Inference-graph propagation [6]	−	○	−	○	−	−
Shapley / SAGE [7]	✓	✓	−	−	−	−
LOCO / permutation	✓	✓	−	−	−	−
MCI [9]	✓	✓	✓	✓	−	−
$M_{i \rightarrow y}^{(K)}$ (this paper)	✓	✓	✓	✓	✓	✓

	Pearson	MI	Single field	LOCO	Permutation	SAGE	Graph	$M^{(2)}$
unit output P_g	0.02	0.03	0.06	0.21	0.86	0.21	0.08	0.21
own-bus LMP	0.00	0.04	0.00	0.02	0.57	0.00	0.05	0.15
nearby LMP (306)	0.00	0.00	0.00	0.01	0.00	0.00	0.05	0.14
other-area LMP (223)	0.00	0.00	0.00	0.00	0.00	0.00	0.04	0.12
remote LMP (303)	0.00	0.01	0.00	0.00	0.00	0.00	0.02	0.10
competitor output (316)	0.00	0.00	0.00	0.00	0.00	0.00	0.03	0.09
system load	0.00	0.01	0.00	0.01	0.01	0.00	0.01	0.05
PV available	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.03
area-3 load	0.00	0.00	0.00	0.01	0.00	0.00	0.01	0.03
wind available	0.00	0.01	0.00	0.00	0.00	0.00	0.01	0.02
congestion C6	0.00	0.00	0.00	-0.01	0.00	0.00	0.00	0.02

Figure G.14: Field scores in the controlled private-markup case. Colors are normalized within each method; printed values use the original scale. Price fields have almost no single-field scores but appreciable budgeted marginal risk.

Appendix H. Minimal Unsafe Sets and Withholding Counts

Table H.13 complements the field counts in Table 2. Its residual counts quantify the combinations missed by withholding only fields that are unsafe individually.

Table H.13: Minimal unsafe sets of size at most three and withholding requirements. MUS: number of minimal unsafe sets; residual: MUS remaining after withholding all fields unsafe alone; hit: minimum number of fields whose withholding breaks every MUS.

Target	$\tau = 0.5$			$\tau = 0.7$		
	MUS	residual	hit	MUS	residual	hit
RTS unit 223	60	50	15	16	6	11
RTS unit 221	52	42	15	34	24	13
RTS line C35	123	119	13	82	82	7
NEM BW01	266	266	4	0	0	0
NEM TUMUT3	23	23	1	0	0	0
NEM PPCCGT	215	213	6	8	8	1
PJM load	237	237	9	44	44	3
PJM total gen.	179	177	11	39	37	5
PJM net interch.	102	102	3	1	1	1
CAISO load	542	537	17	335	332	10

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